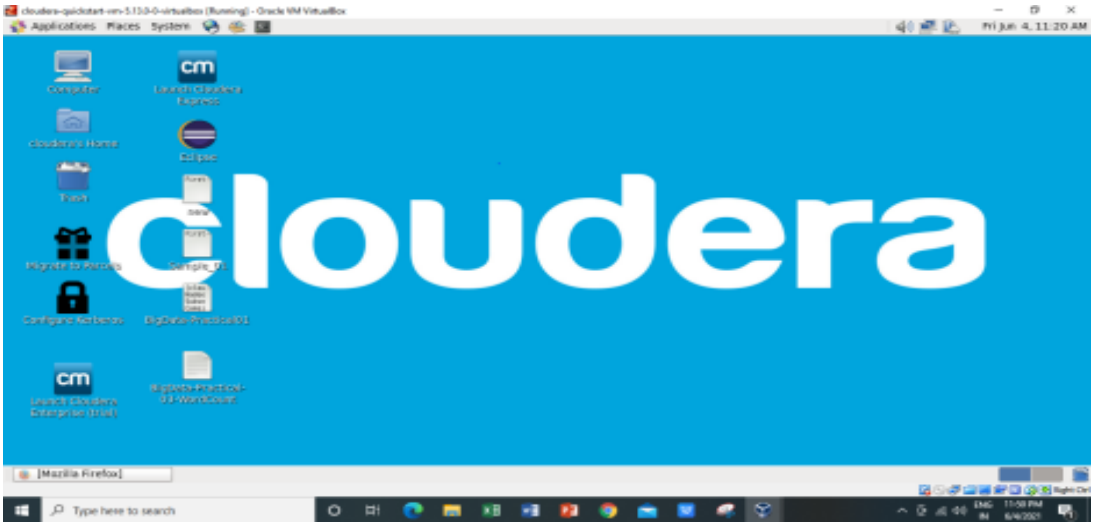
Name:
Roll No:

Steps for Word Count in Cloudera: (Without Combiner)

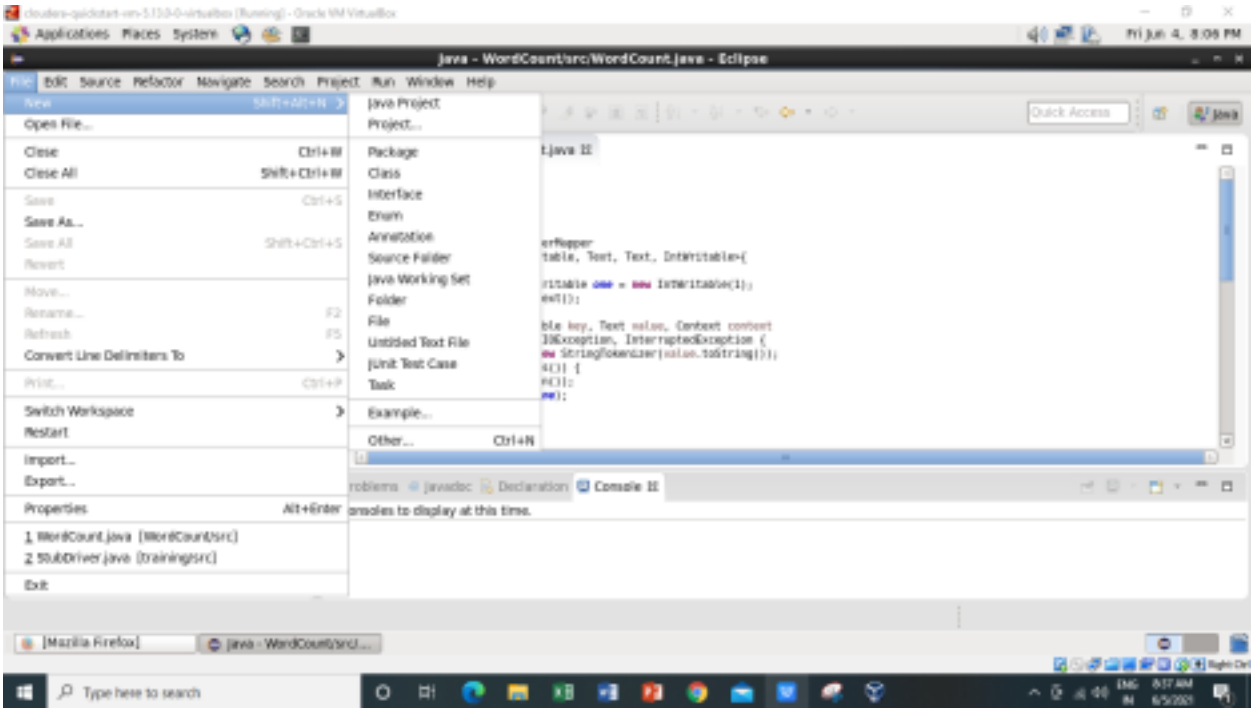
1) Open virtual box and then start cloudera quickstart.



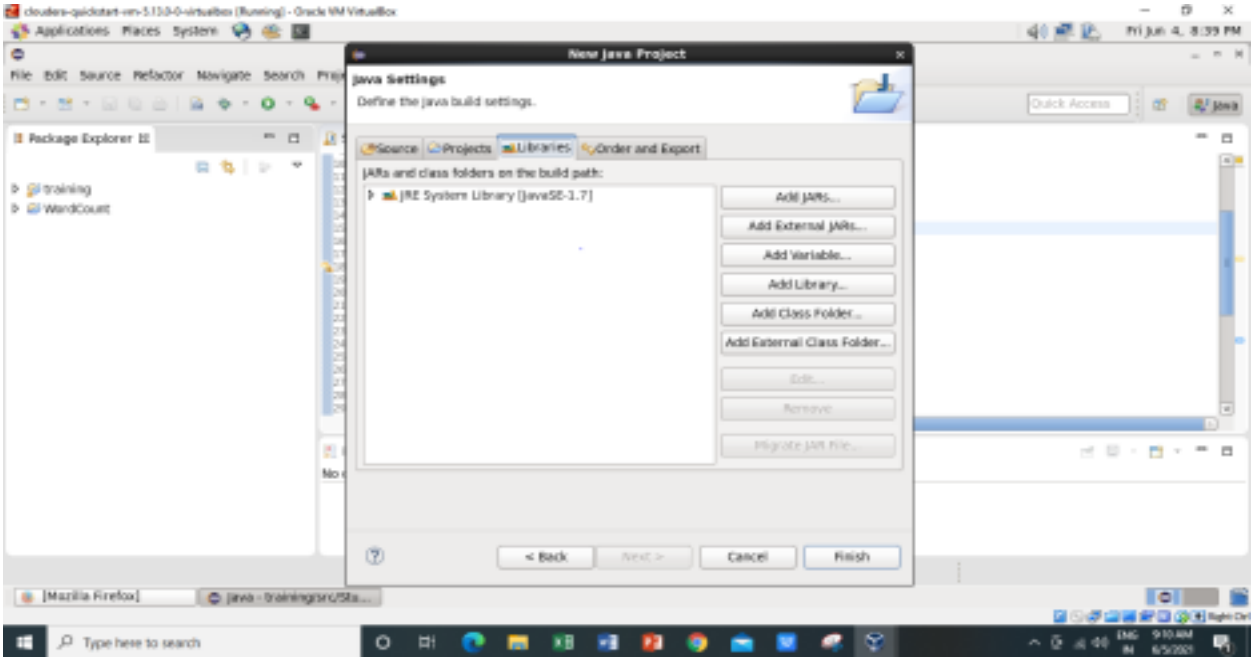
2) Open Eclipse present on the cloudera desktop.

3) Create a new Java project clicking: File -> New -> Project -> Java Project -> Next ("WordCount" is the project name).

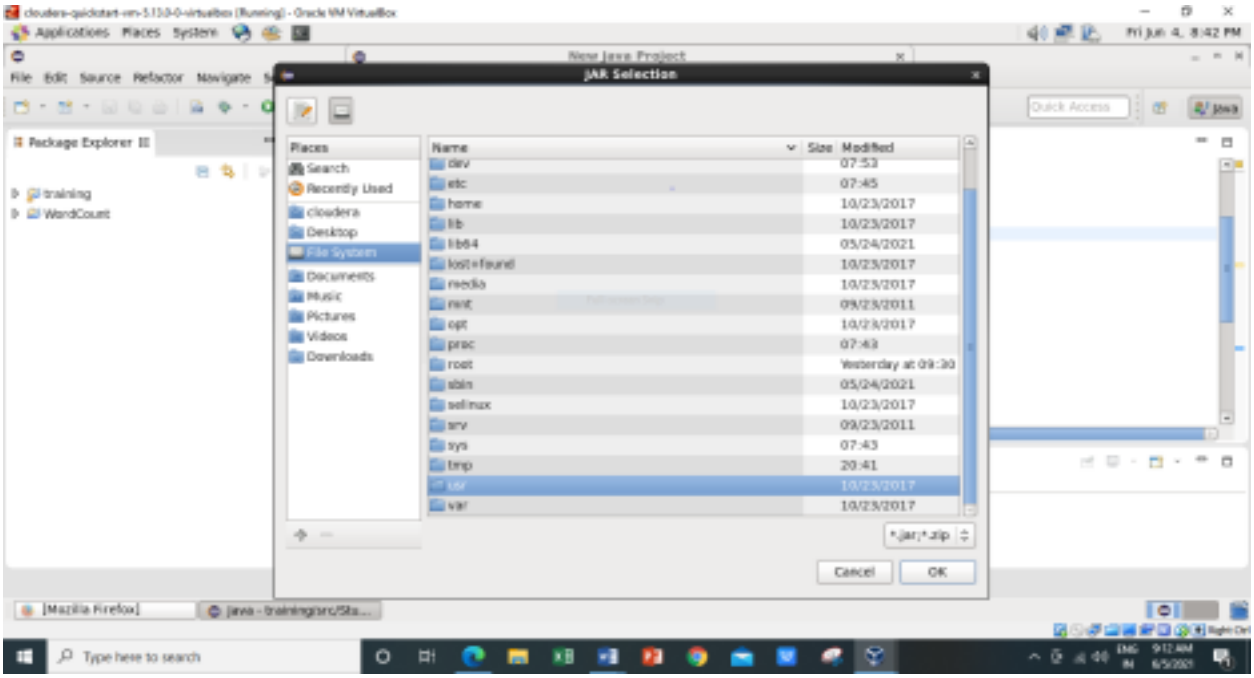
Name:
Roll No:

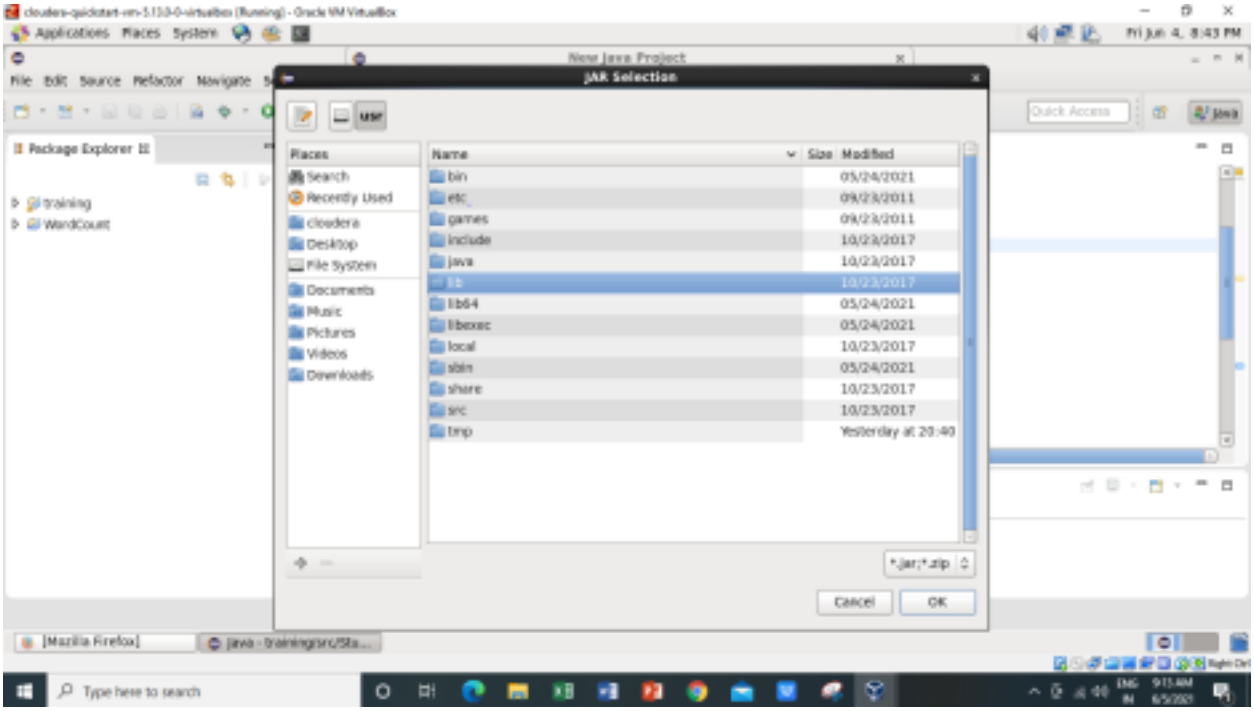


4)Adding the Hadoop libraries to the project Click on Libraries -> Add External JARs Click on File System -> usr -> lib -> hadoop Select all the libraries (JAR Files) -> click OK Click on Add External jars, -> client -> select all jar files -> ok -> Finish

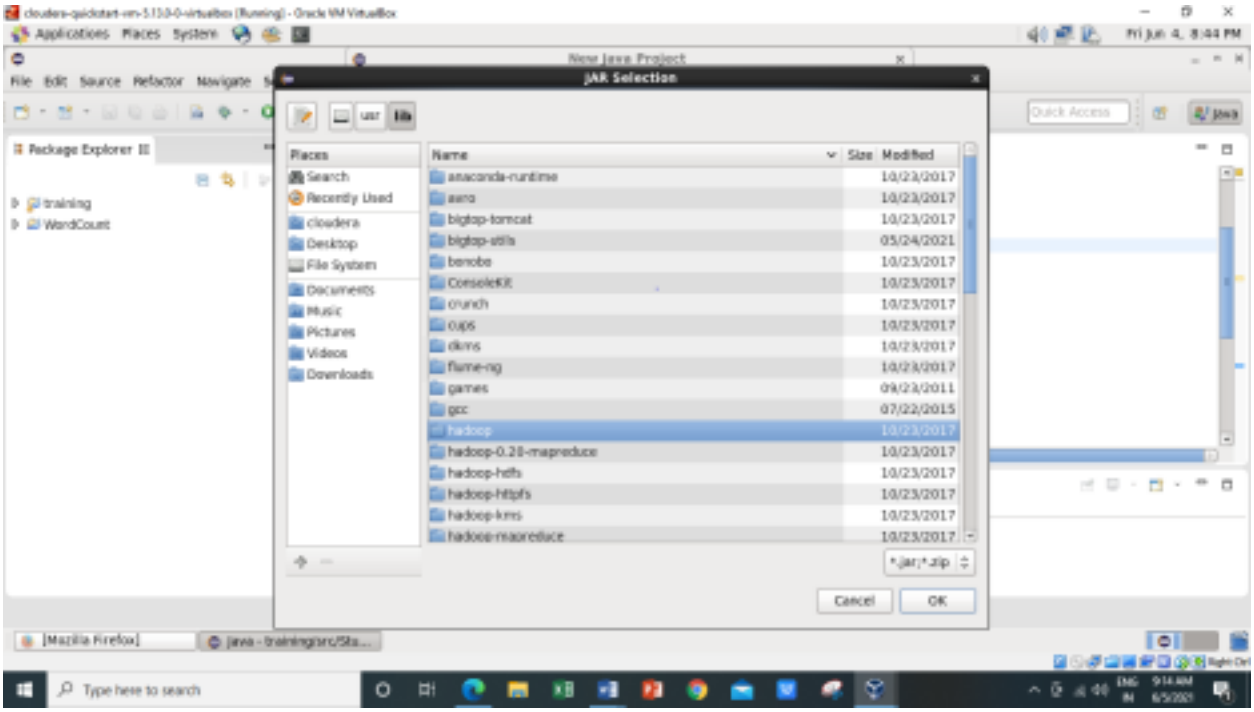


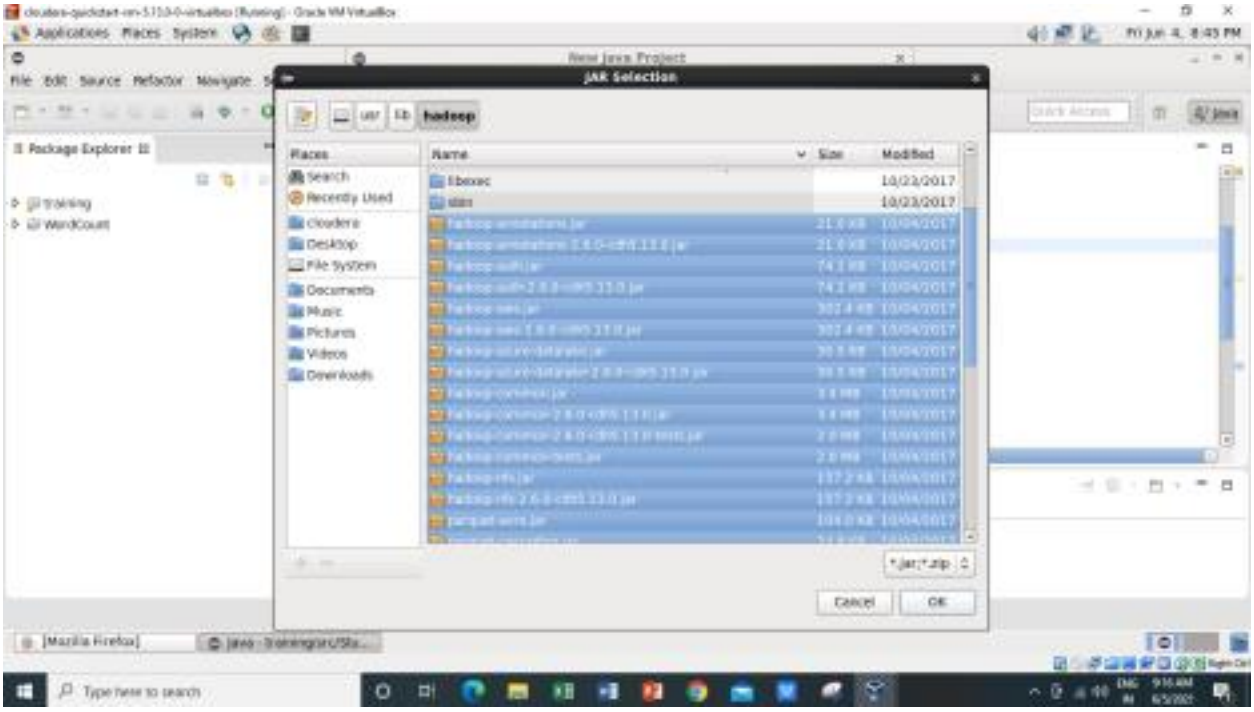
Name:
Roll No:


JAR Selection
Places
Search
Recently Used
cloudera
Desktop
File System
Documents
Music
Pictures
Videos
Downloads
Name
Size
Modified
Cancel
OK

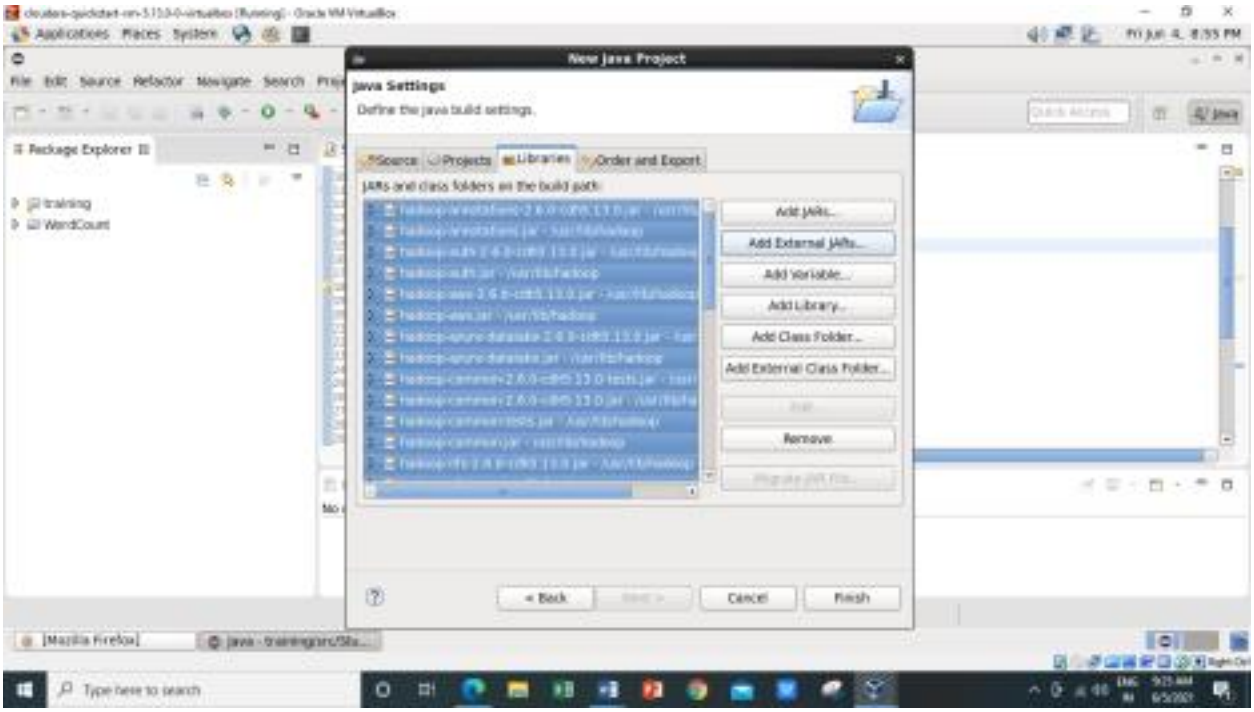

JAR Selection
Places
Search
Recently Used
cloudera
Desktop
File System
Documents
Music
Pictures
Videos
Downloads
Name
Size
Modified
Cancel
OK

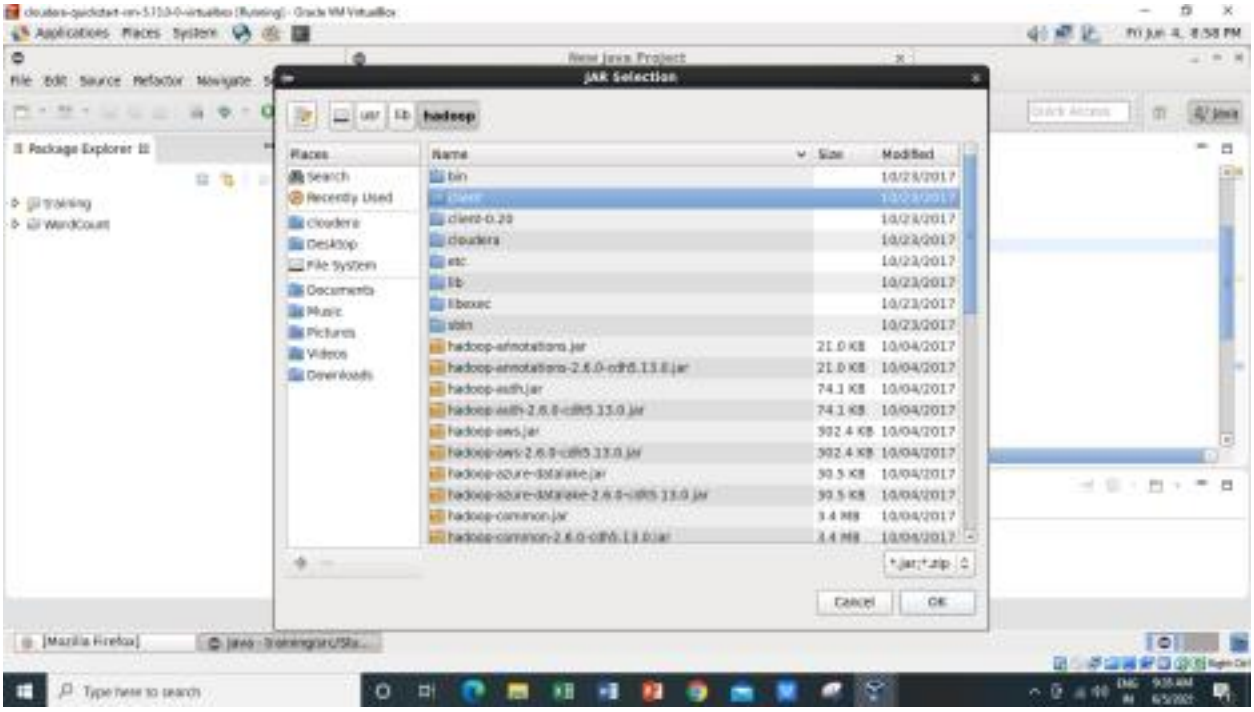
Name:
Roll No:


JAR Selection
New Java Project
Package Explorer
training
WordCount
Places
Search
Recently Used
cloudera
Desktop
File System
Documents
Music
Pictures
Videos
Downloads
Name
Size
Modified
anaconda-runtime
avro
bigtop-tomcat
bigtop-utils
bonobo
ConsoleKit
crunch
cups
dkms
flume-ng
games
gcc
hadoop
hadoop-0.20-mapreduce
hadoop-hdfs
hadoop-httpfs
hadoop-kms
hadoop-mapreduce
.jar;.zip
Cancel
OK

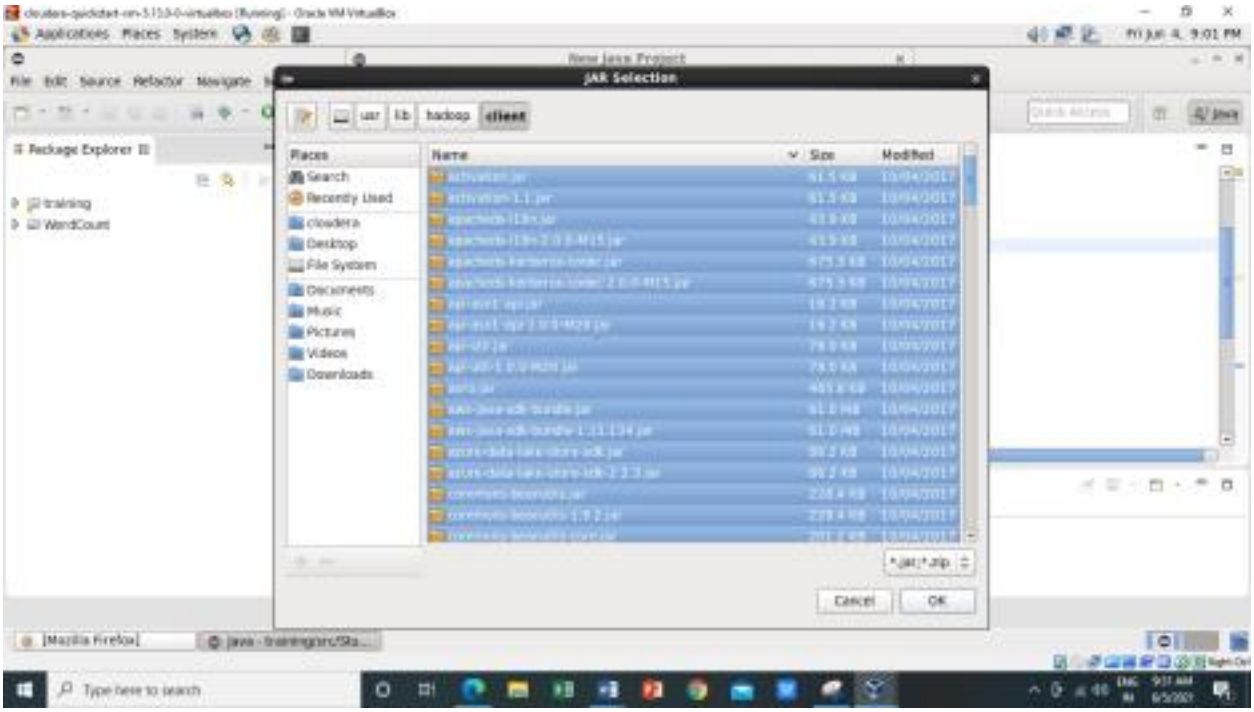

JAR Selection
New Java Project
hadoop
Package Explorer
training
WordCount
Places
Search
Recently Used
cloudera
Desktop
File System
Documents
Music
Pictures
Videos
Downloads
Name
Size
Modified
libexec
sbin
.jar;.zip
Cancel
OK

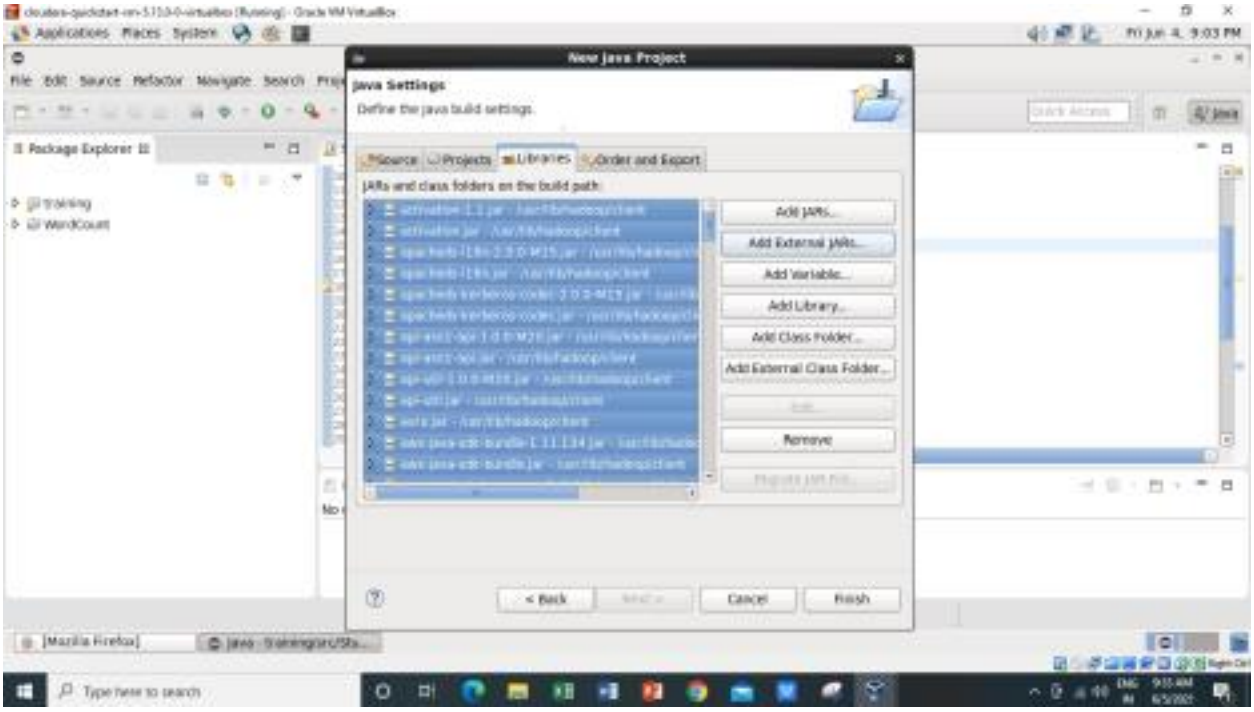
Name:
Roll No:


New Java Project
Java Settings
Define the java build settings.
Source
Projects
Libraries
Order and Export
JARs and class folders on the build path:
Add JARs...
Add External JARs...
Add Variable...
Add Library...
Add Class Folder...
Add External Class Folder...
Edit
Remove
Migrate JAR File...
< Back
Next >
Cancel
Finish


JAR Selection
usr
lib
hadoop
Places
Search
Recently Used
Cloudera
Desktop
File System
Documents
Music
Pictures
Videos
Downloads
Name
Size
Modified
bin
client
client-0.20
cloudera
etc
lib
libexec
sbin
hadoop-annotations.jar
hadoop-annotations-2.6.0-cdh5.13.0.jar
hadoop-auth.jar
hadoop-auth-2.6.0-cdh5.13.0.jar
hadoop-aws.jar
hadoop-aws-2.6.0-cdh5.13.0.jar
hadoop-azure-datalake.jar
hadoop-azure-datalake-2.6.0-cdh5.13.0.jar
hadoop-common.jar
hadoop-common-2.6.0-cdh5.13.0.jar
.jar;.zip
Cancel
OK

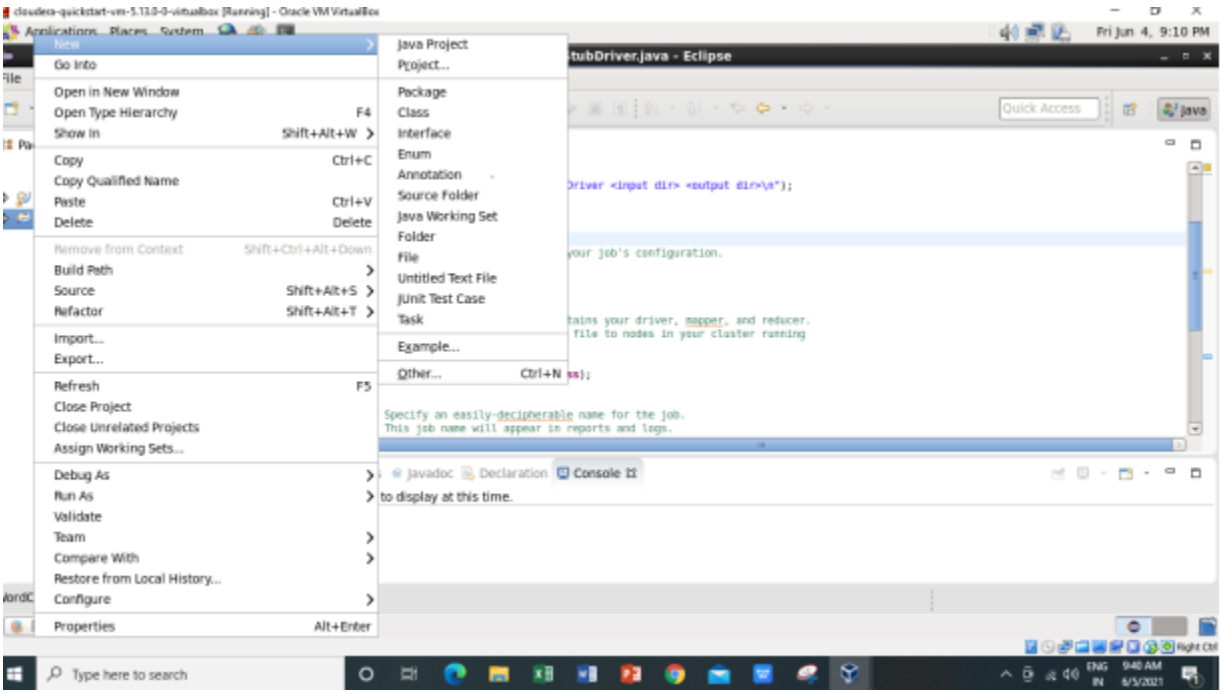
Name:
Roll No:

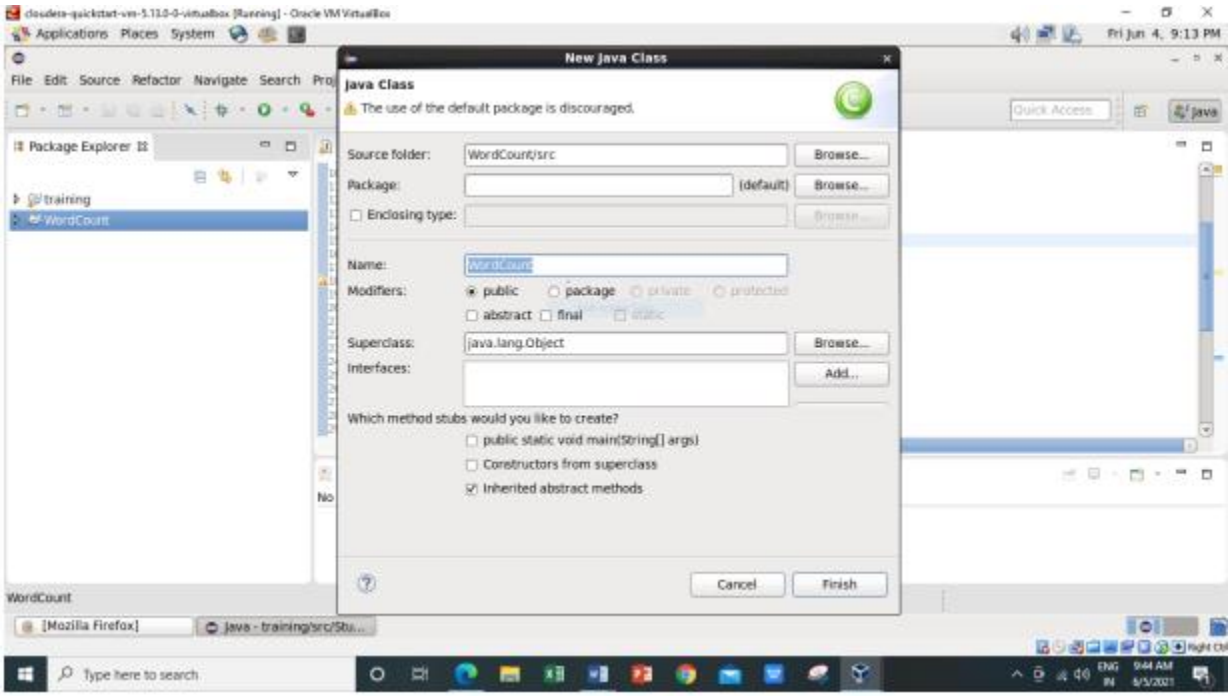




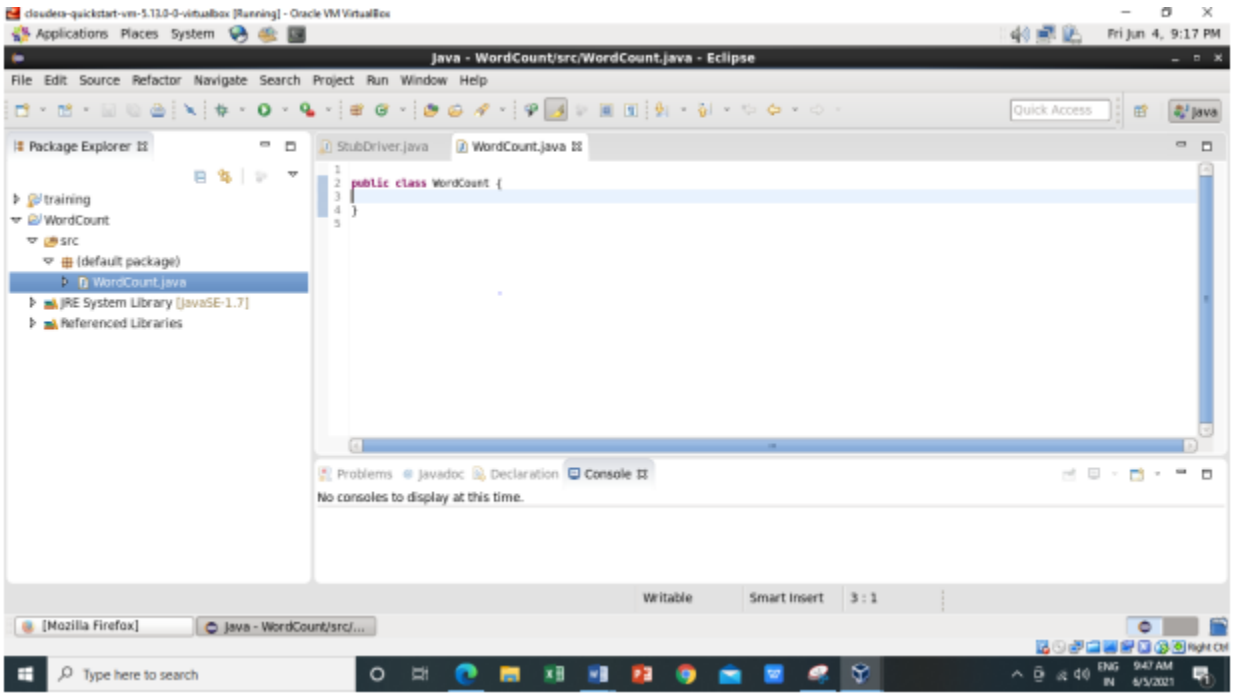
5) Right Click on the name of Project “WordCount” -> New -> class Don’t write anything for package Write Name Textbox write “WordCount” -> Finish Then WordCount.java window will pop up

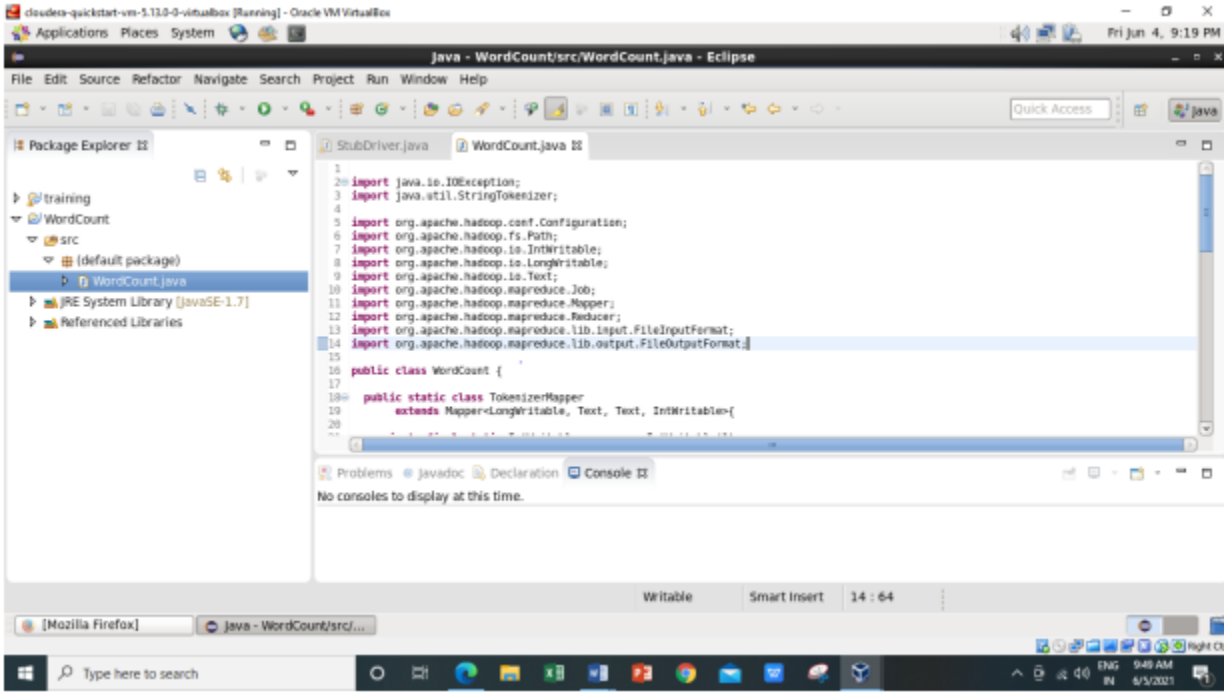
Name:
Roll No:


New
Go Into
Open in New Window
Open Type Hierarchy F4
Show In Shift+Alt+W
Copy Ctrl+C
Copy Qualified Name
Paste Ctrl+V
Delete Delete
Remove from Context Shift+Ctrl+Alt+Down
Build Path
Source Shift+Alt+S
Refactor Shift+Alt+T
Import...
Export...
Refresh F5
Close Project
Close Unrelated Projects
Assign Working Sets...
Debug As
Run As
Validate
Team
Compare With
Restore from Local History...
Configure
Properties Alt+Enter
Java Project
Project...
Package
Class
Interface
Enum
Annotation
Source Folder
Java Working Set
Folder
File
Untitled Text File
JUnit Test Case
Task
Example...
Other... Ctrl+N
Specify an easily-decipherable name for the job.
This job name will appear in reports and logs.
Javadoc
Declaration
Console
to display at this time.
Type here to search


New Java Class
Java Class
The use of the default package is discouraged.
Source folder: WordCount/src
Browse...
Package:
(default)
Enclosing type:
Name: WordCount
Modifiers: public package private protected
abstract final static
Superclass: java.lang.Object
Interfaces:
Add...
Which method stubs would you like to create?
public static void main(String[] args)
Constructors from superclass
Inherited abstract methods
Cancel
Finish
Package Explorer
training
WordCount
File Edit Source Refactor Navigate Search
Type here to search

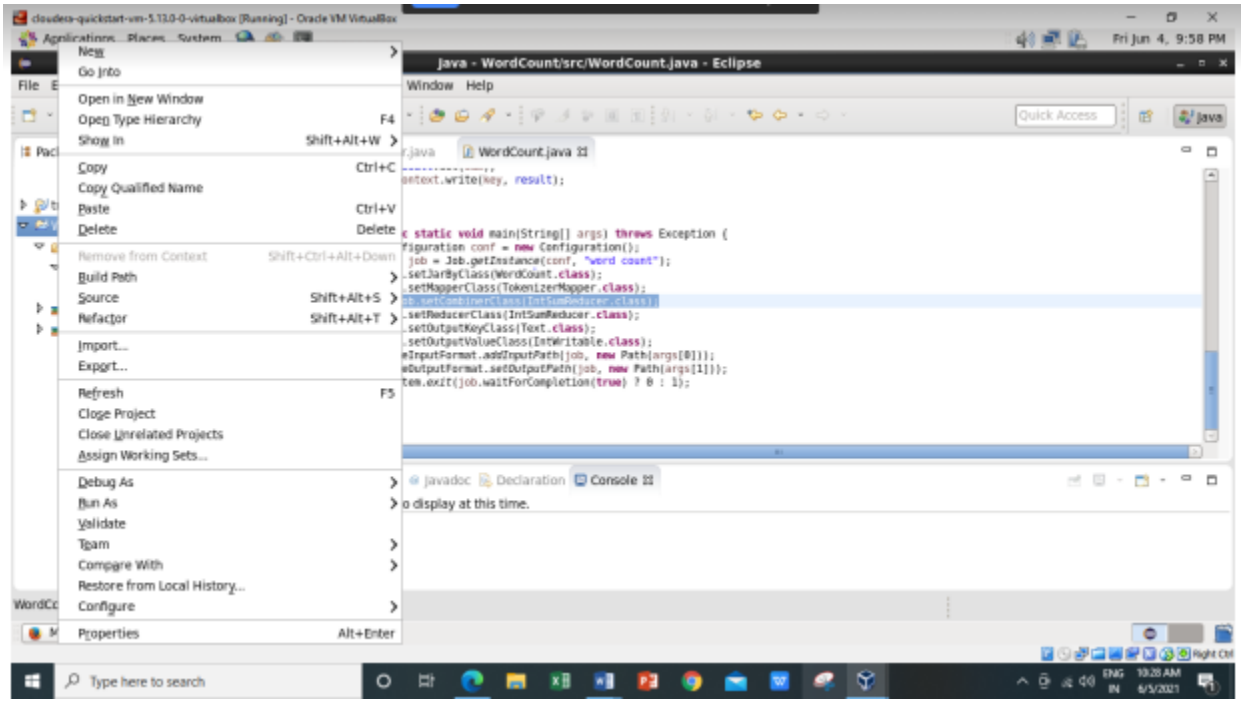
Name:
Roll No:

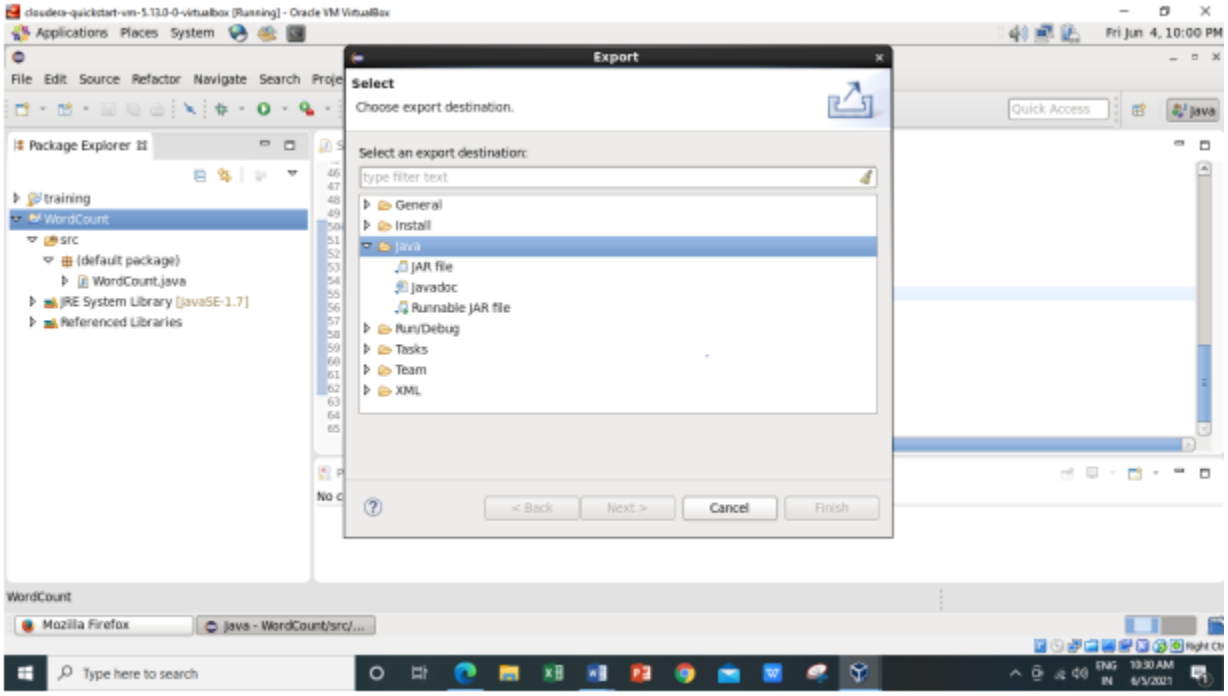

Java - WordCount/src/WordCount.java - Eclipse
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer
training
WordCount
src
(default package)
WordCount.java
JRE System Library [javaSE-1.7]
Referenced Libraries
StubDriver.java
WordCount.java
public class WordCount {
}
Problems Javadoc Declaration Console
No consoles to display at this time.
Writable
Smart Insert
3 : 1


Java - WordCount/src/WordCount.java - Eclipse
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer
training
WordCount
src
(default package)
WordCount.java
JRE System Library [javaSE-1.7]
Referenced Libraries
StubDriver.java
WordCount.java
import java.io.IOException;
import java.util.StringTokenizer;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
public class WordCount {
public static class TokenizerMapper
extends Mapper<LongWritable, Text, Text, IntWritable>{
Problems Javadoc Declaration Console
No consoles to display at this time.
Writable
Smart Insert
14 : 64

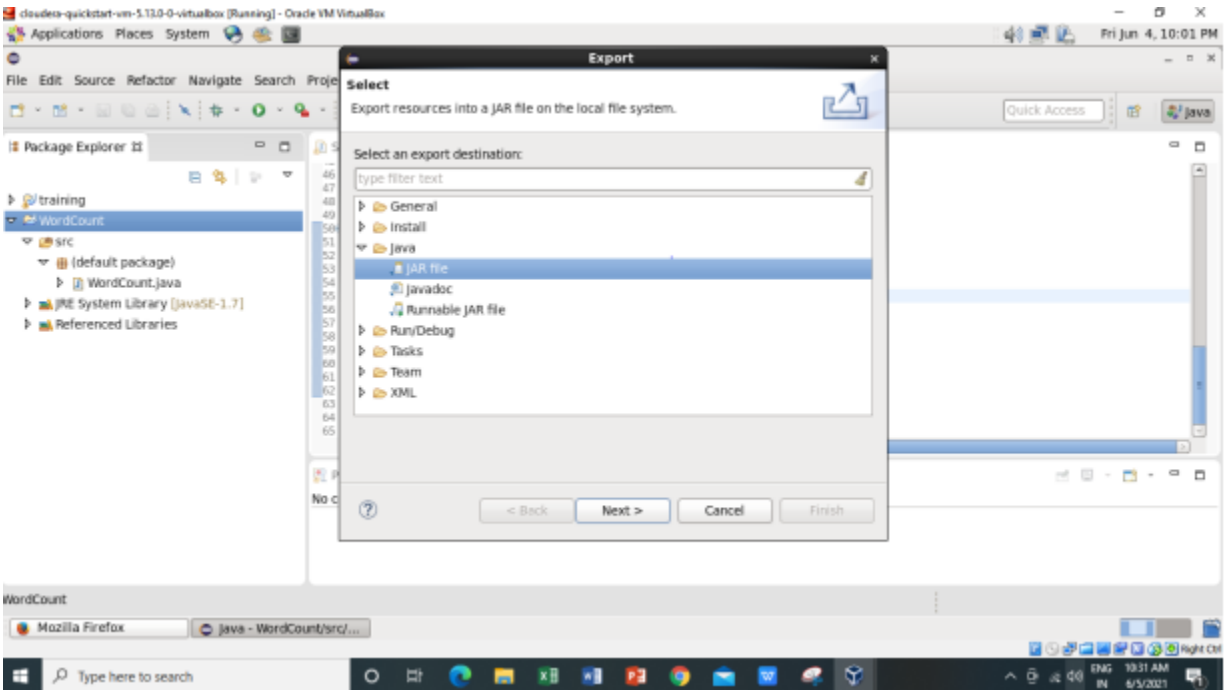
Name:
Roll No:

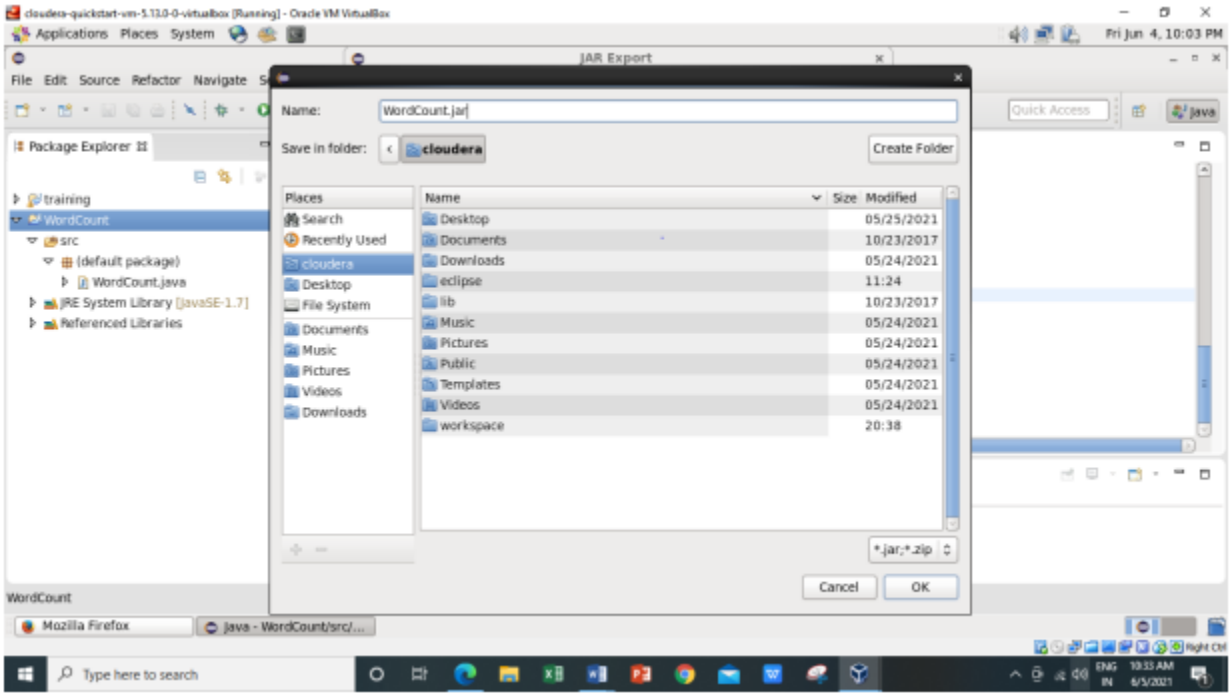
6) Right Click on the project name WordCount -> Export -> Java -> JAR File -> Next -> for select the export destination for JAR file: browse -> Name : WordCount.jar -> save in folder -> cloudera -> Finish -> OK



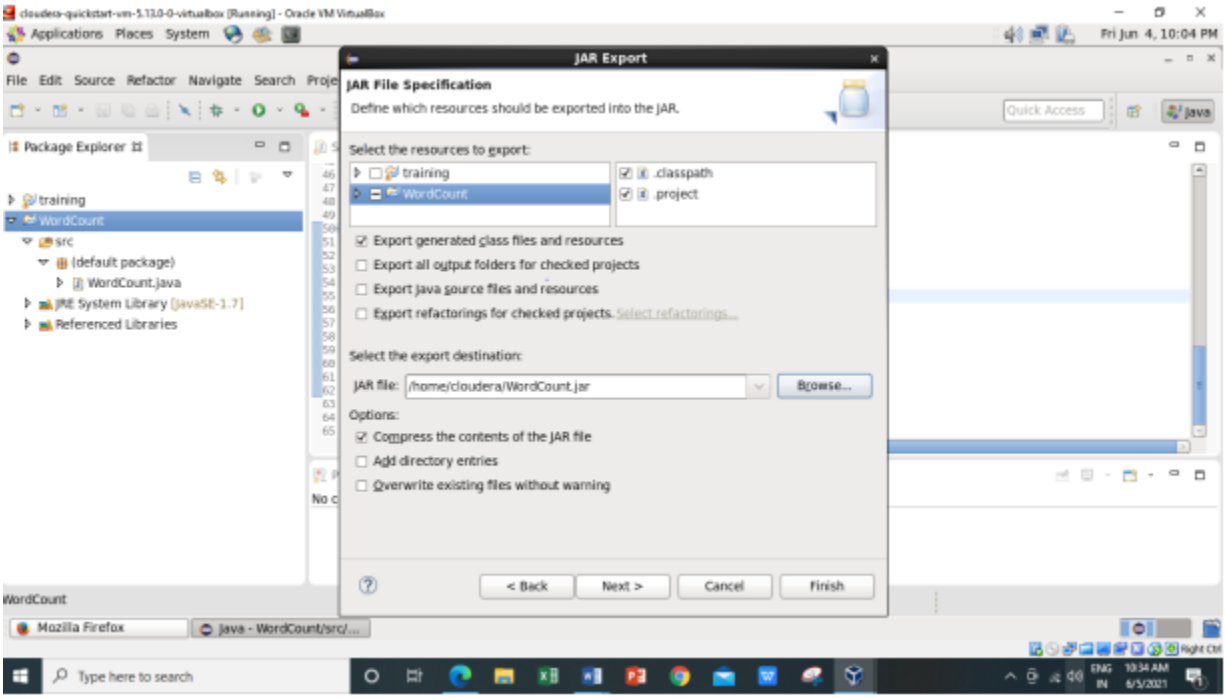


Name:
Roll No:


Export
Select
Export resources into a JAR file on the local file system.
Select an export destination:
type filter text
General
Install
Java
JAR file
Javadoc
Runnable JAR file
Run/Debug
Tasks
Team
XML
< Back
Next >
Cancel
Finish
Package Explorer
training
WordCount
src
(default package)
WordCount.java
JRE System Library [JavaSE-1.7]
Referenced Libraries


JAR Export
Name:
WordCount.jar
Save in folder:
cloudera
Create Folder
Places
Search
Recently Used
cloudera
Desktop
File System
Documents
Music
Pictures
Videos
Downloads
Name
Size
Modified
Desktop 05/25/2021
Documents 10/23/2017
Downloads 05/24/2021
eclipse 11:24
lib 10/23/2017
Music 05/24/2021
Pictures 05/24/2021
Public 05/24/2021
Templates 05/24/2021
Videos 05/24/2021
workspace 20:38
.jar;.zip
Cancel
OK

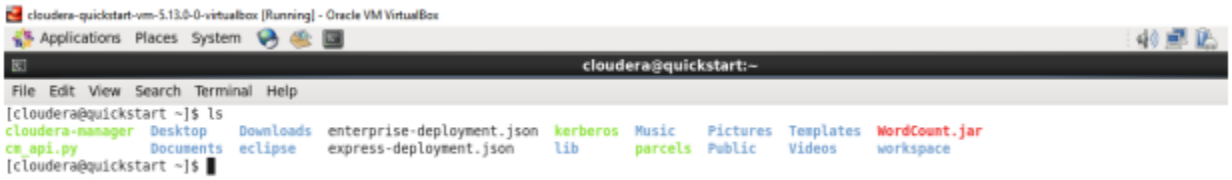
Name:
Roll No:



7) Verify jar file from terminal by using Open terminal & type " ls " There it will show WordCount.jar

Check current working directory ->pwd

```
[cloudera@quickstart ~]$ pwd
/home/cloudera
[cloudera@quickstart ~]$
```



8) We need to create an input file in local file system

Creating an input file named as "abc".

Name:
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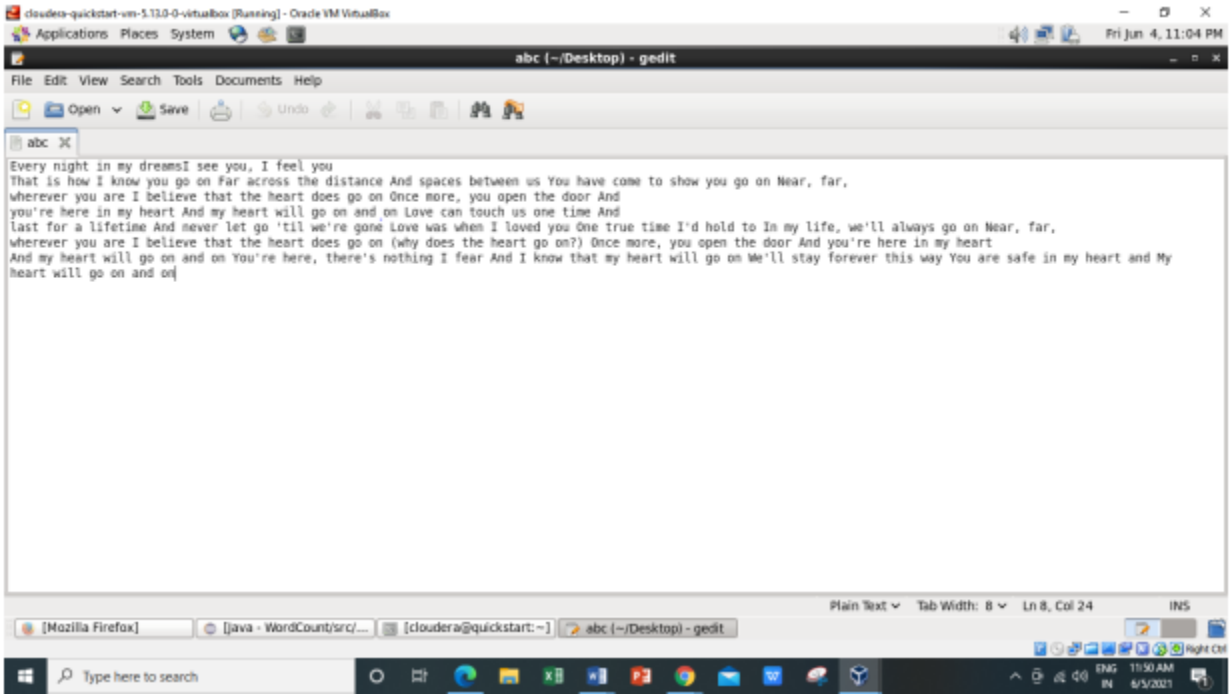


Here listing all the directory present in hdfs using hdfs dfs -ls / command

```
[cloudera@quickstart ~]$ hdfs dfs -ls /
Found 10 items
-rw-r--r--   1 cloudera supergroup         27 2021-05-24 12:04 /Sample_01
drwxrwxrwx   - hdfs     supergroup          0 2017-10-23 09:15 /benchmarks
drwxr-xr-x   - cloudera supergroup          0 2021-05-24 13:58 /forcopy
drwxr-xr-x   - hbase    supergroup          0 2021-06-04 07:57 /hbase
drwxr-xr-x   - cloudera supergroup          0 2021-05-24 13:20 /newdir
drwxr-xr-x   - cloudera supergroup          0 2021-05-24 13:36 /rjc
drwxr-xr-x   - cloudera supergroup          0 2021-05-24 13:55 /solr
drwxrwxrwt   - hdfs     supergroup          0 2021-05-24 10:39 /tmp
drwxr-xr-x   - hdfs     supergroup          0 2017-10-23 09:17 /user
drwxr-xr-x   - hdfs     supergroup          0 2017-10-23 09:17 /var
[cloudera@quickstart ~]$
```

9) Now we have to move this input file to hdfs. For this we create a direcory on hdfs using command hdfs dfs -mkdir /inputnew.

```
[cloudera@quickstart ~]$ hdfs dfs -mkdir /inputdir
[cloudera@quickstart ~]$
```

Then we can verify whether this directory is created or not using ls command hdfs dfs -ls /

Name:
Roll No:

```
[cloudera@quickstart ~]$ hdfs dfs -ls /
Found 11 items
-rw-r--r--   1 cloudera supergroup         27 2021-05-24 12:04 /Sample_01
drwxrwxrwx   - hdfs     supergroup          0 2017-10-23 09:15 /benchmarks
drwxr-xr-x   - cloudera supergroup          0 2021-05-24 13:58 /forcopy
drwxr-xr-x   - hbase    supergroup          0 2021-06-04 07:57 /hbase
drwxr-xr-x   - cloudera supergroup          0 2021-06-04 23:34 /inputdir
drwxr-xr-x   - cloudera supergroup          0 2021-05-24 13:20 /newdir
drwxr-xr-x   - cloudera supergroup          0 2021-05-24 13:36 /rjc
drwxr-xr-x   - cloudera supergroup          0 2021-05-24 13:55 /solr
drwxrwxrwt   - hdfs     supergroup          0 2021-05-24 10:39 /tmp
drwxr-xr-x   - hdfs     supergroup          0 2017-10-23 09:17 /user
drwxr-xr-x   - hdfs     supergroup          0 2017-10-23 09:17 /var
[cloudera@quickstart ~]$
```

Move the input file to this directory created in hdfs by using either put command or copyFromLocal command.

```
[cloudera@quickstart ~]$ hdfs dfs -put /home/cloudera/Desktop/abc /inputdir/
[cloudera@quickstart ~]$
```

Now checking whether the “abc” present in /inputdir directory of hdfs or not using hdfs dfs -ls /inputdir command

```
[cloudera@quickstart ~]$ hdfs dfs -ls /inputdir
Found 1 items
-rw-r--r--   1 cloudera supergroup        813 2021-06-05 00:06 /inputdir/abc
[cloudera@quickstart ~]$
```

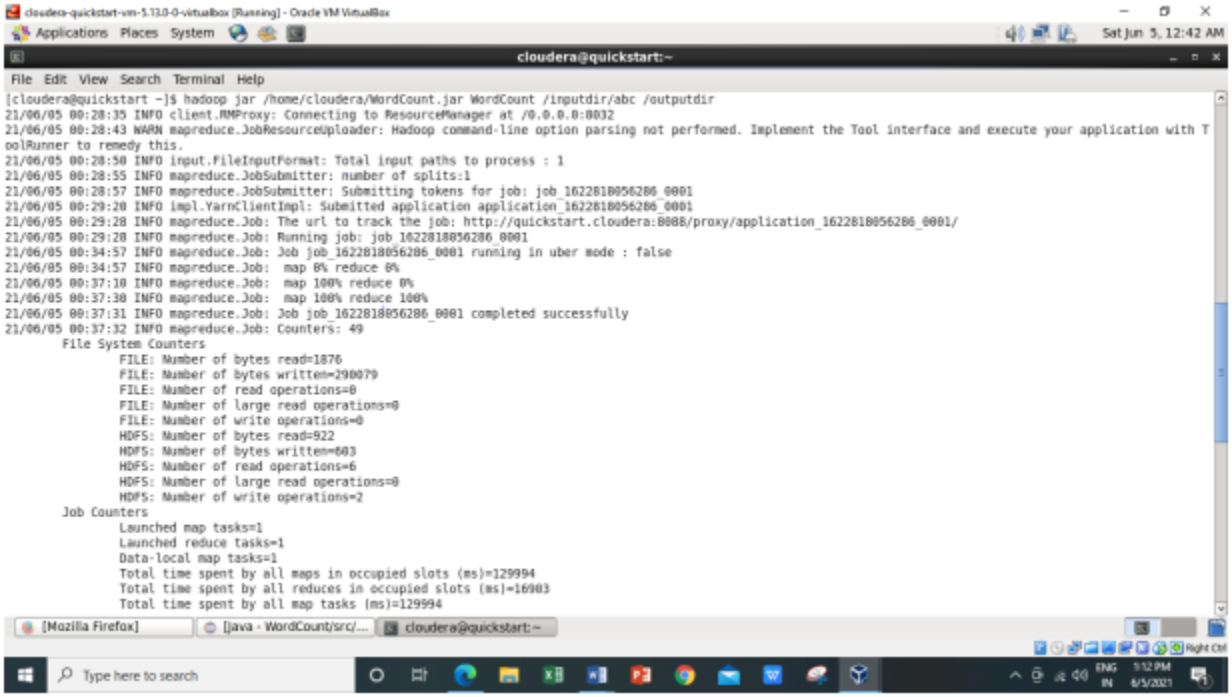
As we can see “abc” file is present in /inputdir directory of hdfs. Now we will see the content of this file using hdfs dfs –cat /inputdir/abc command

```
[cloudera@quickstart ~]$ hdfs dfs -cat /inputdir/abc
Every night in my dreamsI see you, I feel you
That is how I know you go on Far across the distance And spaces between us You have come to show you go on Near, far,
wherever you are I believe that the heart does go on Once more, you open the door And
you're here in my heart And my heart will go on and on Love can touch us one time And
last for a lifetime And never let go 'til we're gone Love was when I loved you One true time I'd hold to In my life, we'll always go on Near, far,
wherever you are I believe that the heart does go on (why does the heart go on?) Once more, you open the door And you're here in my heart
And my heart will go on and on You're here, there's nothing I fear And I know that my heart will go on We'll stay forever this way You are safe in my heart and My
heart will go on and on
[cloudera@quickstart ~]$
```

10) Running Mapreduce Program on Hadoop, syntax is hadoop jar jarFileName.jar ClassName /InputFileAddress /outputdir

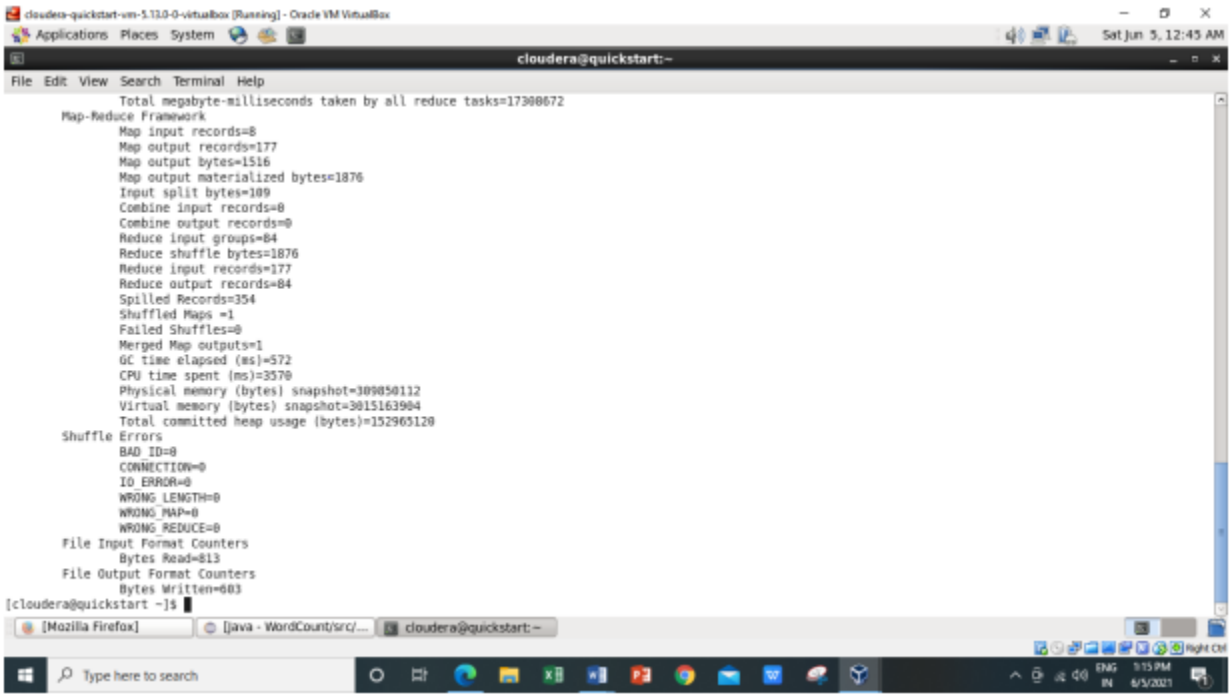
i.e. hadoop jar /home/cloudera/WordCount.jar WordCount /inputdir/abc /outputdir

Name:
Roll No:



```
[cloudera@quickstart ~]$ hadoop jar /home/cloudera/WordCount.jar WordCount /inputdir/abc /outputdir
21/06/05 00:28:35 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
21/06/05 00:28:43 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with T
oolRunner to remedy this.
21/06/05 00:28:50 INFO input.FileInputFormat: Total input paths to process : 1
21/06/05 00:28:55 INFO mapreduce.JobSubmitter: number of splits:1
21/06/05 00:28:57 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1622818056286_0001
21/06/05 00:29:20 INFO impl.YarnClientImpl: Submitted application application_1622818056286_0001
21/06/05 00:29:28 INFO mapreduce.Job: The url to track the job: http://quickstart.cloudera:8088/proxy/application_1622818056286_0001/
21/06/05 00:29:28 INFO mapreduce.Job: Running job: job_1622818056286_0001
21/06/05 00:34:57 INFO mapreduce.Job: Job job_1622818056286_0001 running in uber mode : false
21/06/05 00:34:57 INFO mapreduce.Job:  map 0% reduce 0%
21/06/05 00:37:10 INFO mapreduce.Job:  map 100% reduce 0%
21/06/05 00:37:30 INFO mapreduce.Job:  map 100% reduce 100%
21/06/05 00:37:31 INFO mapreduce.Job: Job job_1622818056286_0001 completed successfully
21/06/05 00:37:32 INFO mapreduce.Job: Counters: 49
        File System Counters
                FILE: Number of bytes read=1876
                FILE: Number of bytes written=290079
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=922
                HDFS: Number of bytes written=603
                HDFS: Number of read operations=6
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
        Job Counters
                Launched map tasks=1
                Launched reduce tasks=1
                Data-local map tasks=1
                Total time spent by all maps in occupied slots (ms)=129994
                Total time spent by all reduces in occupied slots (ms)=16903
                Total time spent by all map tasks (ms)=129994
```

Map-Reduce Framework



```
                Total megabyte-milliseconds taken by all reduce tasks=17308672
        Map-Reduce Framework
                Map input records=8
                Map output records=177
                Map output bytes=1516
                Map output materialized bytes=1876
                Input split bytes=109
                Combine input records=0
                Combine output records=0
                Reduce input groups=84
                Reduce shuffle bytes=1876
                Reduce input records=177
                Reduce output records=84
                Spilled Records=354
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=572
                CPU time spent (ms)=3570
                Physical memory (bytes) snapshot=309050112
                Virtual memory (bytes) snapshot=3015163904
                Total committed heap usage (bytes)=152965120
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=813
        File Output Format Counters
                Bytes Written=603
[cloudera@quickstart ~]$
```

As we can see in the above output,

Combine input records=0

Combine output records=0

Name:
Roll No:

We are getting this because we have commented the Combiner line in main function.

And Reduce shuffle bytes coming as,

Reduce shuffle bytes=1876

So when we are not using combiner 1876 bytes acting as an input for the reducer.

11) Then we can verify the content of outputdir directory and in that part-r file has the actual output by using the command Hdfs dfs -cat /outputdir/part-r-00000 This will give us final output. The same file can also be accessed using a browser. For every execution of this program we need to delete the output directory or give a new name to the output directory every time.

1st we are checking whether the outputdir directory is created in hdfs or not using command

hdfs dfs -ls /

```
[cloudera@quickstart ~]$ hdfs dfs -ls /
Found 12 items
-rw-r--r--   1 cloudera supergroup         27 2021-05-24 12:04 /Sample_01
drwxrwxrwx   - hdfs     supergroup          0 2017-10-23 09:15 /benchmarks
drwxr-xr-x   - cloudera supergroup          0 2021-05-24 13:58 /forcopy
drwxr-xr-x   - hbase    supergroup          0 2021-06-04 07:57 /hbase
drwxr-xr-x   - cloudera supergroup          0 2021-06-05 00:06 /inputdir
drwxr-xr-x   - cloudera supergroup          0 2021-05-24 13:20 /newdir
drwxr-xr-x   - cloudera supergroup          0 2021-06-05 00:37 /outputdir
drwxr-xr-x   - cloudera supergroup          0 2021-05-24 13:36 /rjc
drwxr-xr-x   - cloudera supergroup          0 2021-05-24 13:55 /solr
drwxrwxrwt   - hdfs     supergroup          0 2021-05-24 10:39 /tmp
drwxr-xr-x   - hdfs     supergroup          0 2017-10-23 09:17 /user
drwxr-xr-x   - hdfs     supergroup          0 2017-10-23 09:17 /var
[cloudera@quickstart ~]$
```

Now let's check what we have inside this **outputdir** directory using command as **hdfs dfs -ls /outputdir**

```
[cloudera@quickstart ~]$ hdfs dfs -ls /outputdir
Found 2 items
-rw-r--r--   1 cloudera supergroup          0 2021-06-05 00:37 /outputdir/_SUCCESS
-rw-r--r--   1 cloudera supergroup        603 2021-06-05 00:37 /outputdir/part-r-00000
[cloudera@quickstart ~]$
```

Now we want to read the content of the **part-r-00000 file** which present inside the **outputdir** using command **hdfs dfs -cat /outputdir/part-r-00000**

Name:
Roll No:

```
[cloudera@quickstart ~]$ hdfs dfs -cat /outputdir/part-r-00000
'til    1
(why    1
And     8
Every   1
Far     1
I       7
I'd     1
In      1
Love    2
My      1
Near,   2
Once    2
One     1
That    1
We'll   1
You     2
You're  1
a       1
across  1
always  1
and     4
are     3
believe 2
between 1
can     1
come    1
distance        1
does    3
door    2
dreamsI 1
```

Name:
Roll No:

```
more,   2
my      8
never   1
night   1
nothing 1
on      12
on?)    1
one     1
open    2
safe    1
see     1
show    1
spaces  1
stay    1
that    3
the     6
there's 1
this    1
time    2
to      2
touch   1
true    1
us      2
was     1
way     1
we'll   1
we're   1
when    1
wherever        2
will    4
you     8
you're  2
you,    1
[cloudera@quickstart ~]$
```

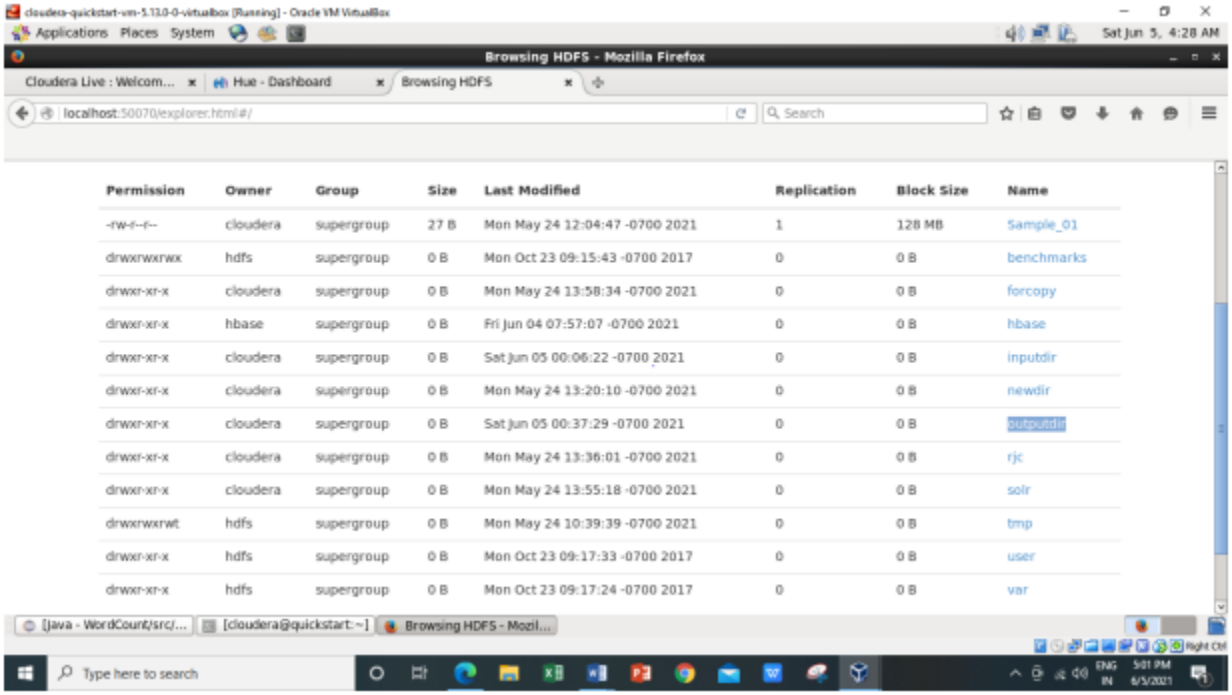
It will give the count of number of times each word has occurred as output.

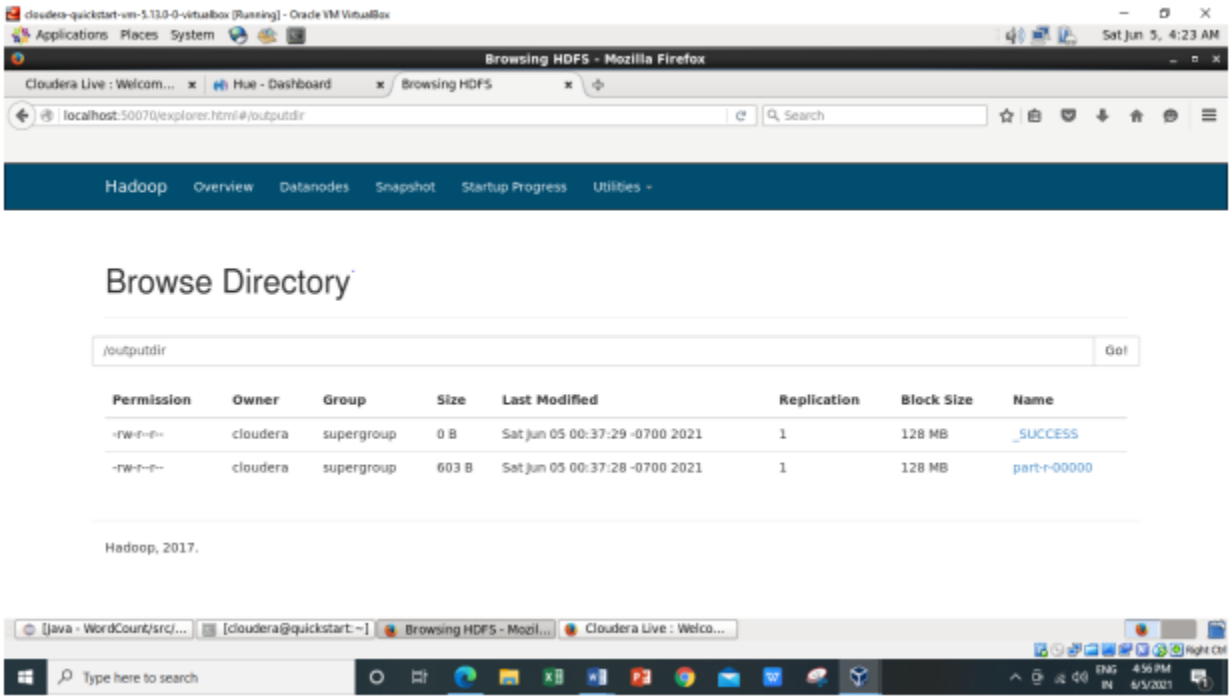
12) The same file can also be accessed using a browser.

Browse the Directory by

Hadoop->HDFS Namenode->Ultilities ->Browse the file system

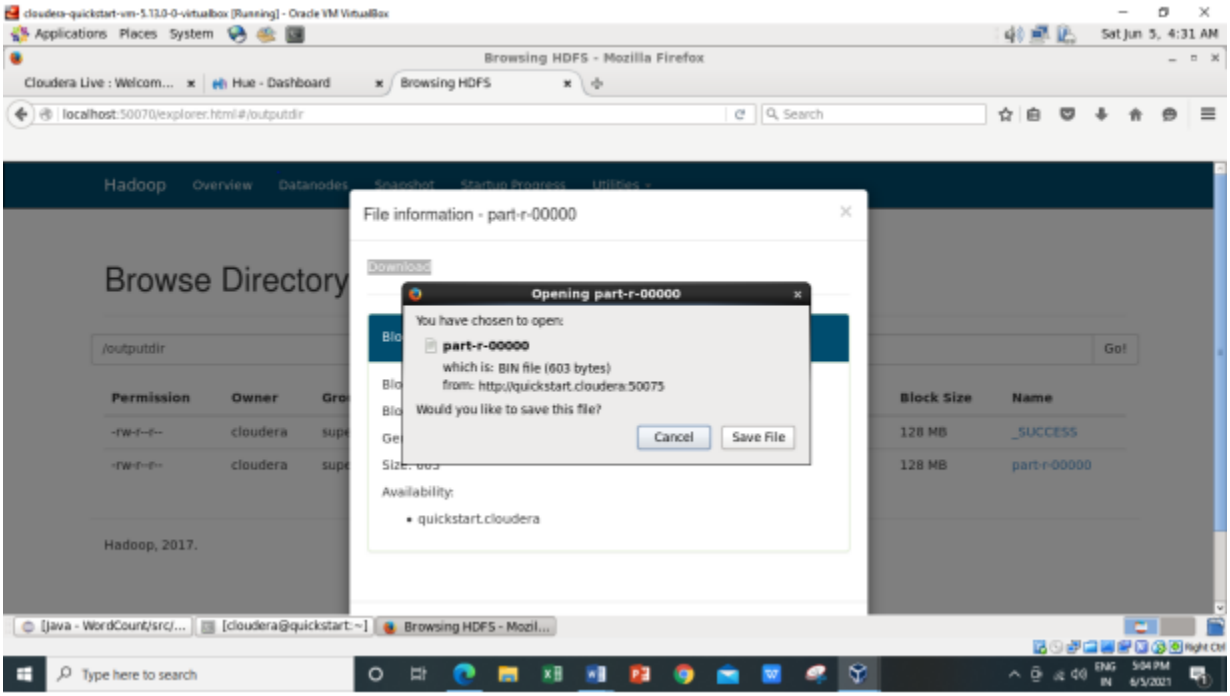
Name:
Roll No:



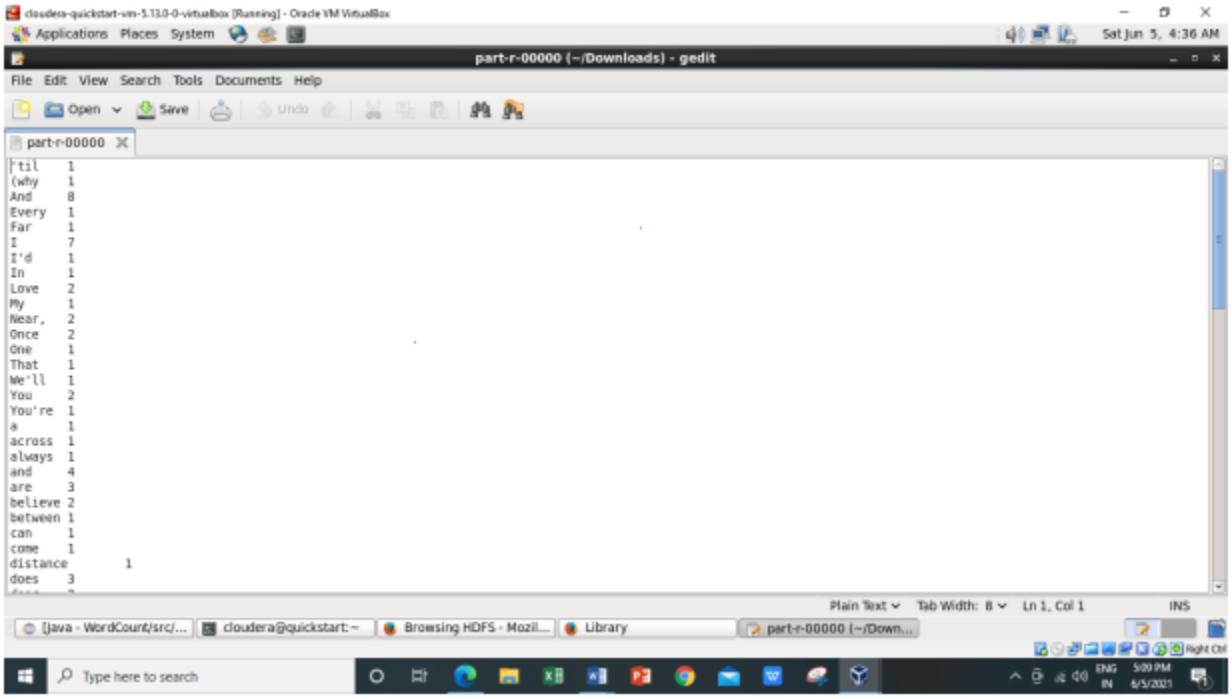


Now downloading the **part-r-00000** file.

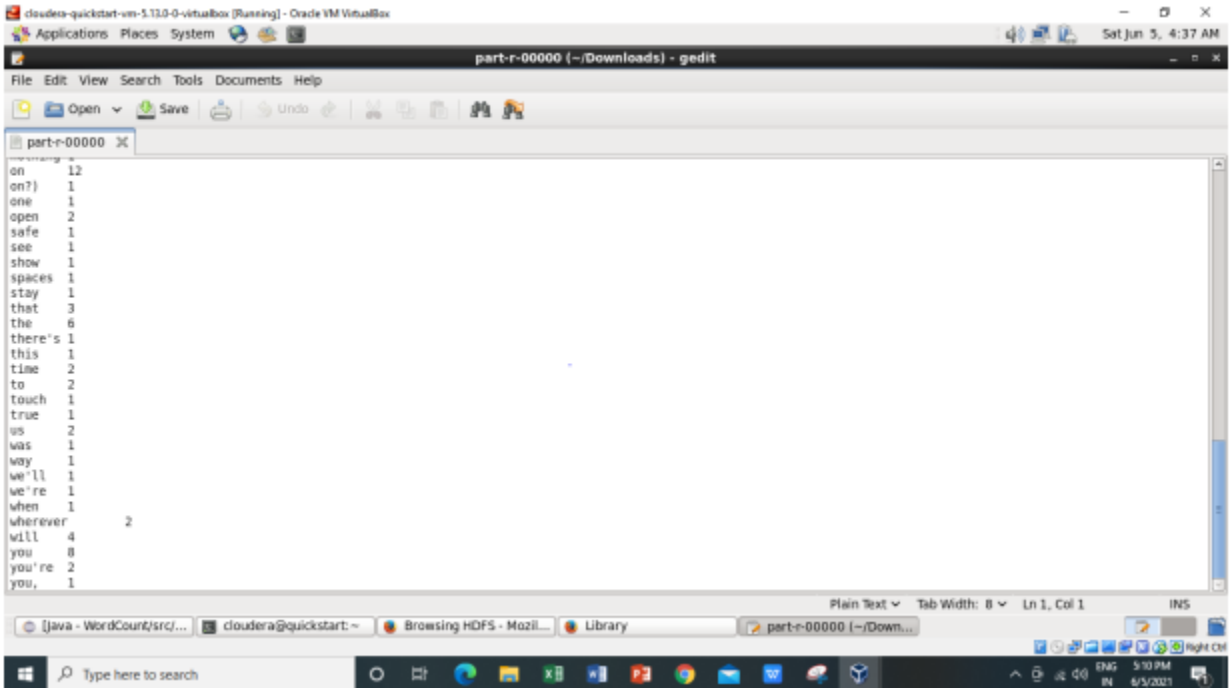
Name:
Roll No:



Inside the **part-r-00000** file it will have the same output as we are getting after executing using command **hadoop jar /home/cloudera/WordCount.jar WordCount /inputdir/abc /op1**



Name:
Roll No:

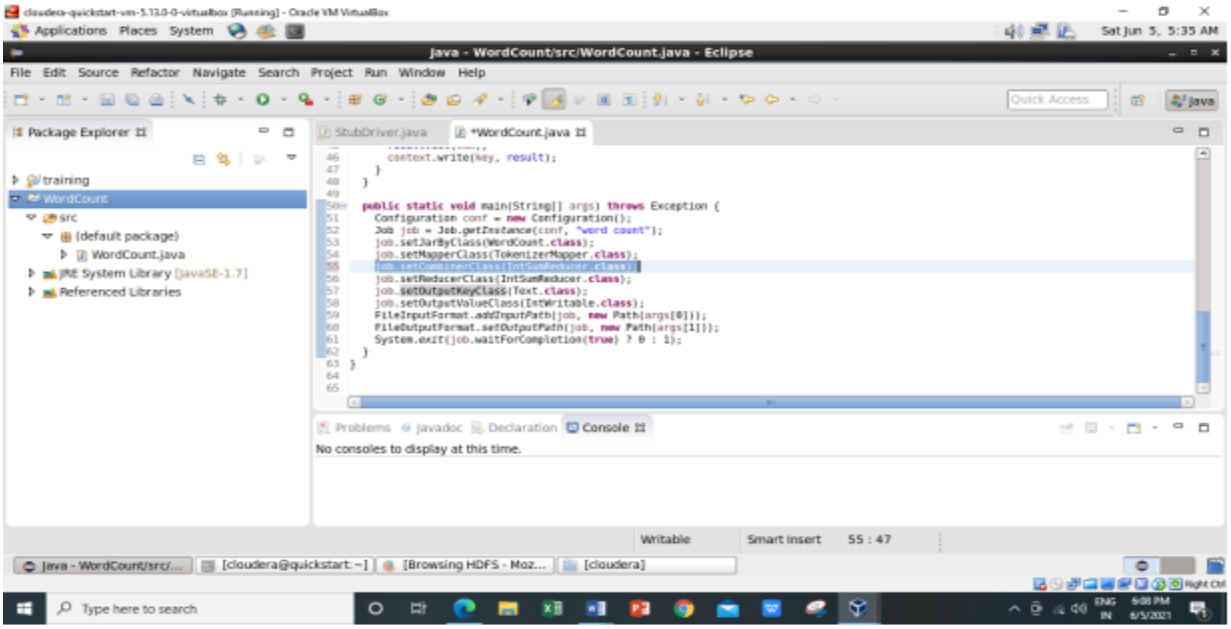


For every execution of this program we need to delete the output directory or give a new name to the output directory every time.

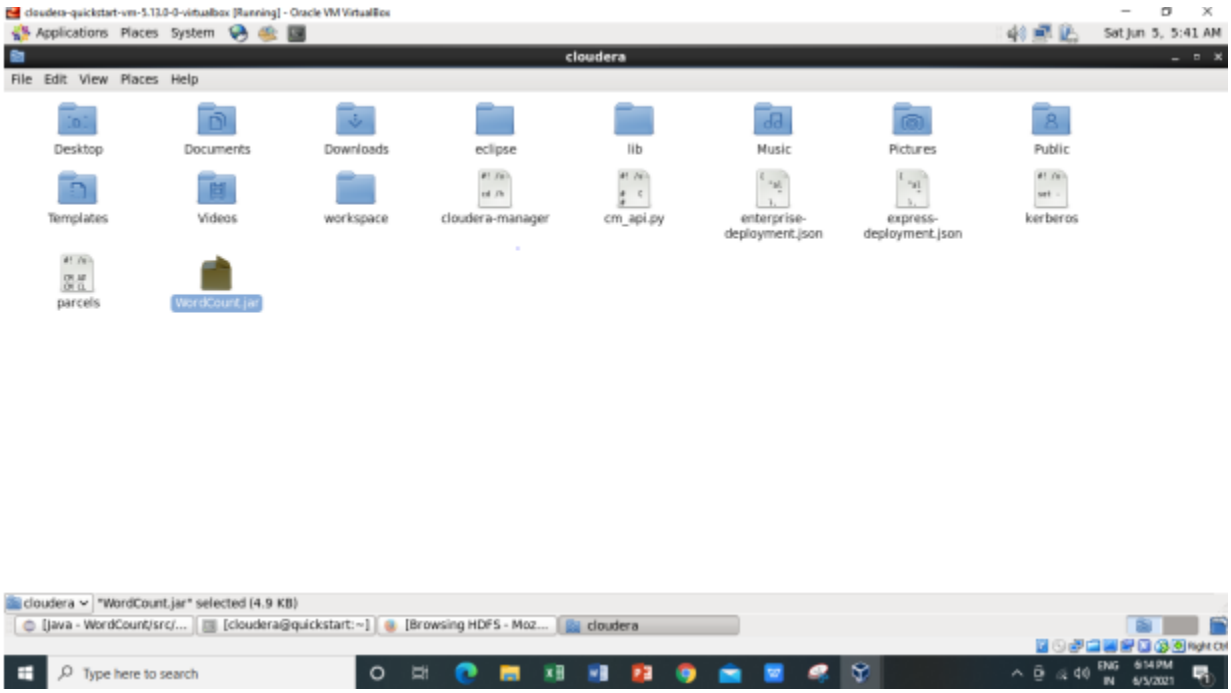
Implementation of WordCount problem using Hadoop MapReduce (With Combiner) in Eclipse:

1) We will perform the same steps as we have done above for WordCount (without using combiner) in that we just uncommenting the combiner line in main function**.**

Name:
Roll No:

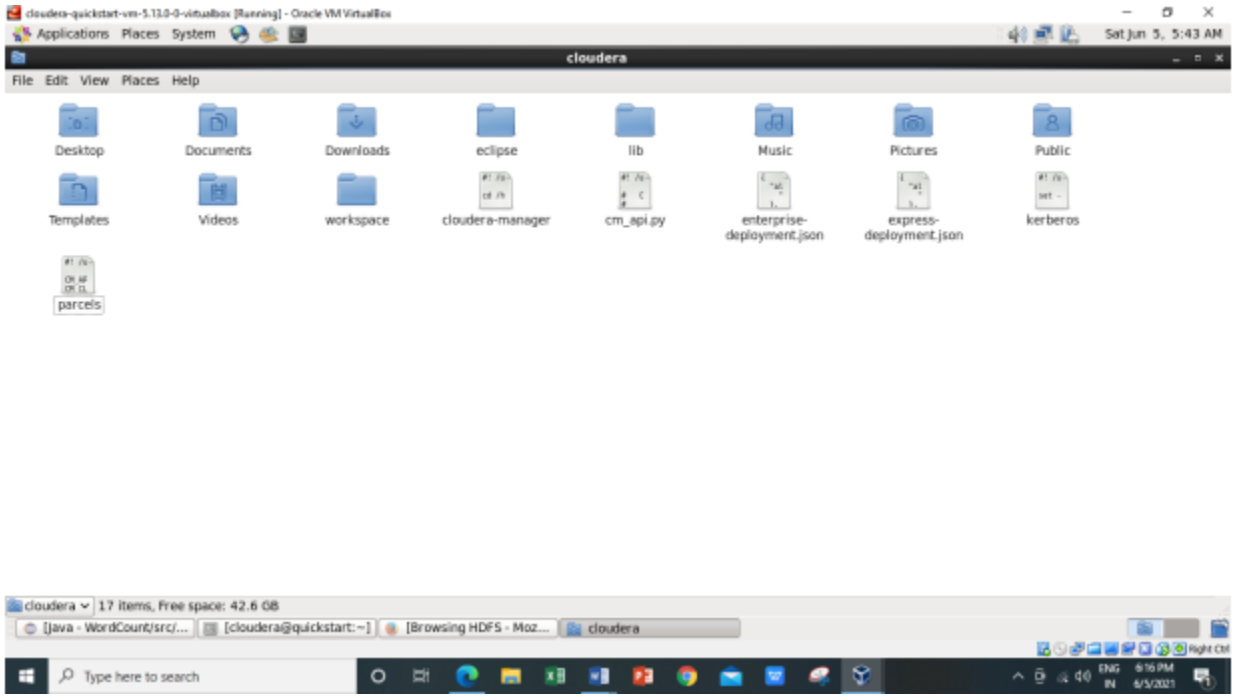


2)And will delete the WordCount.jar file in which all jar files are present from **/home/cloudera.**

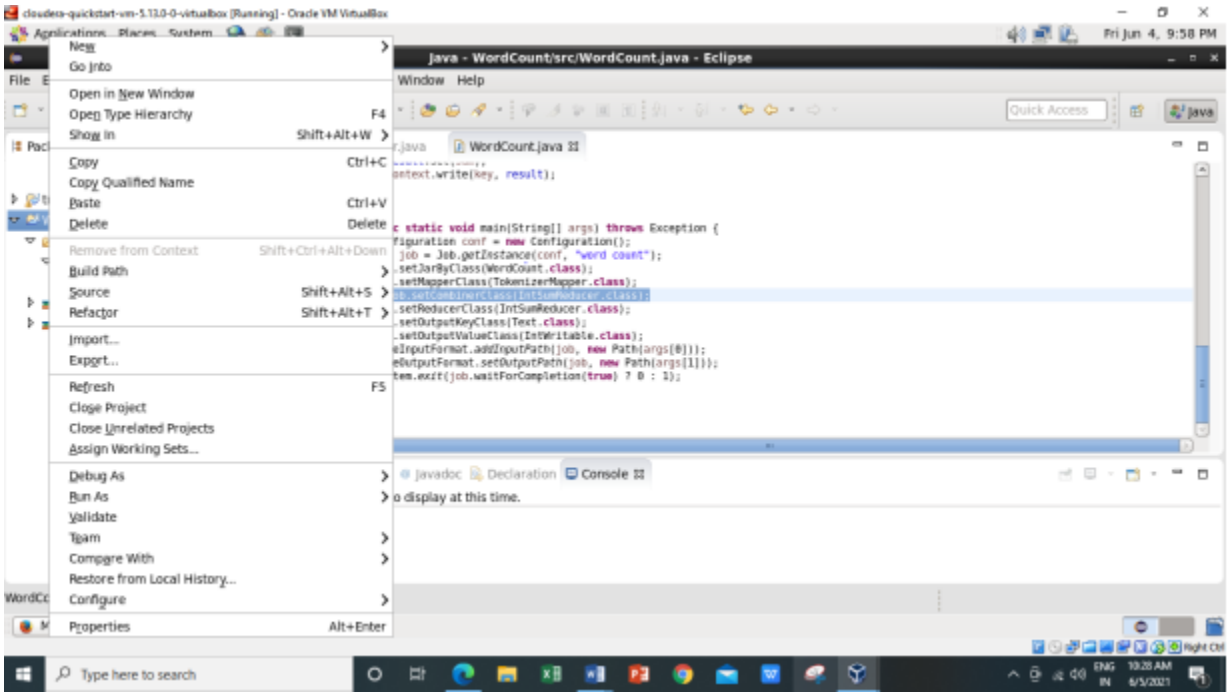


We have successfully deleted the WordCount.jar file.

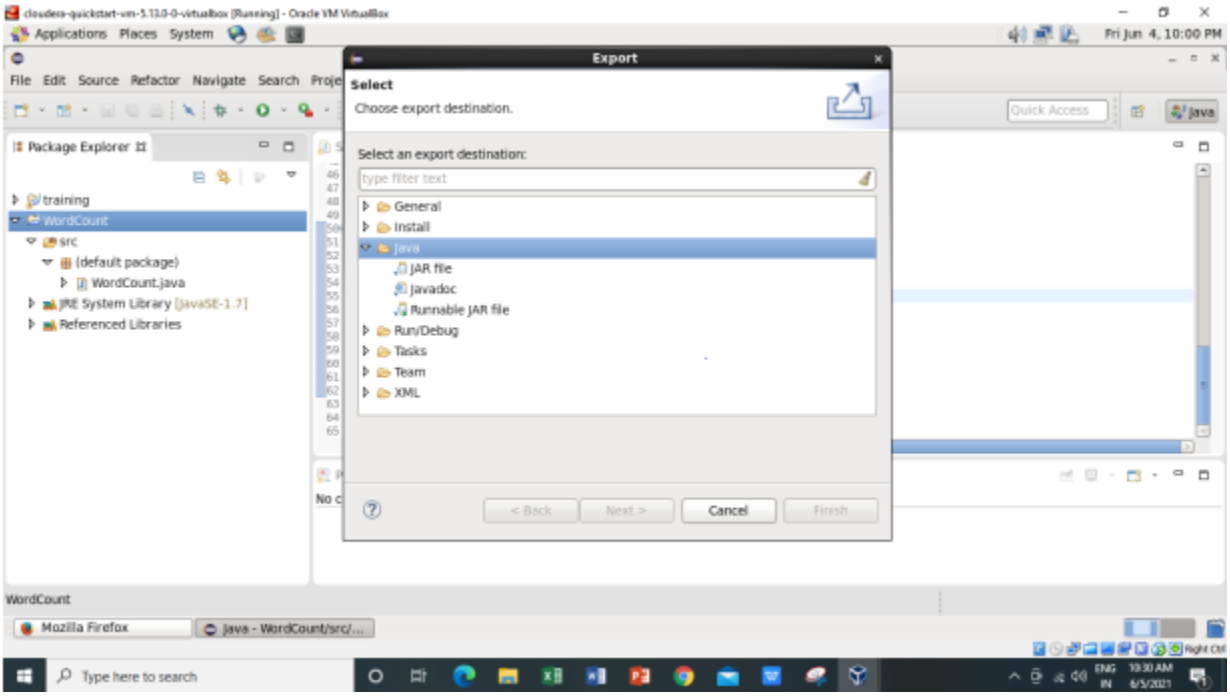
Name:
Roll No:

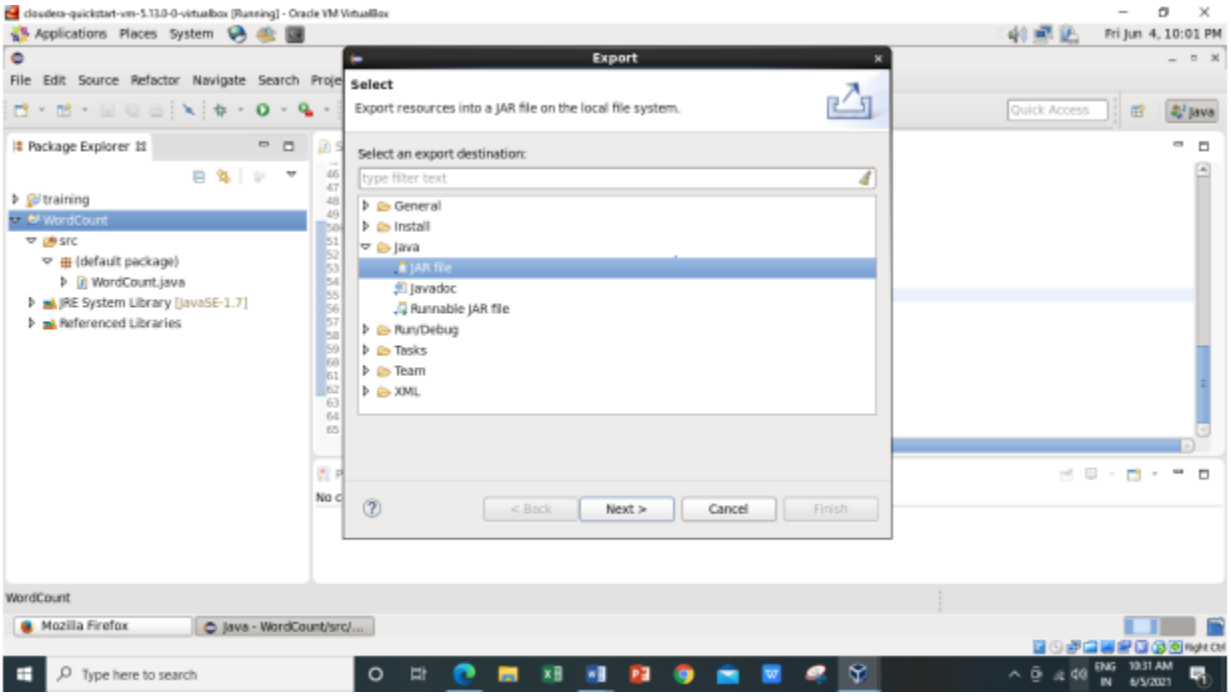


3) Now exporting the jar files Right Click on the project name WordCount -> Export -> Java ->JAR File -> Next -> for select the export destination for JAR file: browse -> Name : WordCount.jar -> save in folder -> cloudera -> Finish -> OK

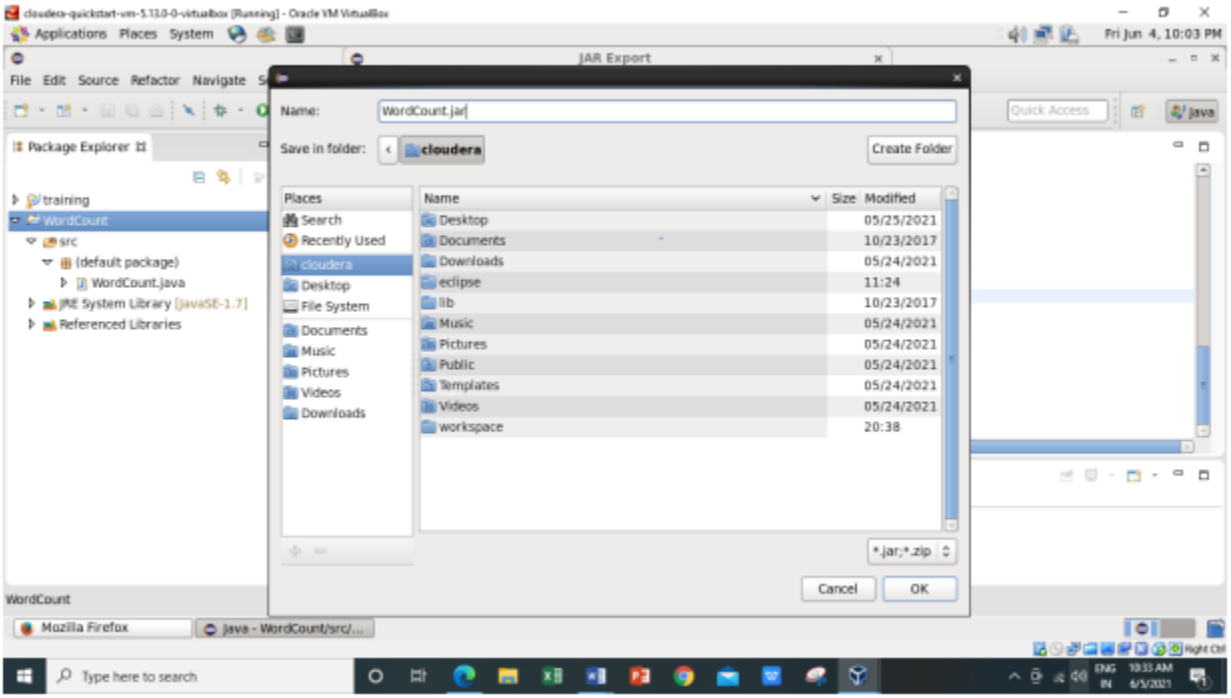


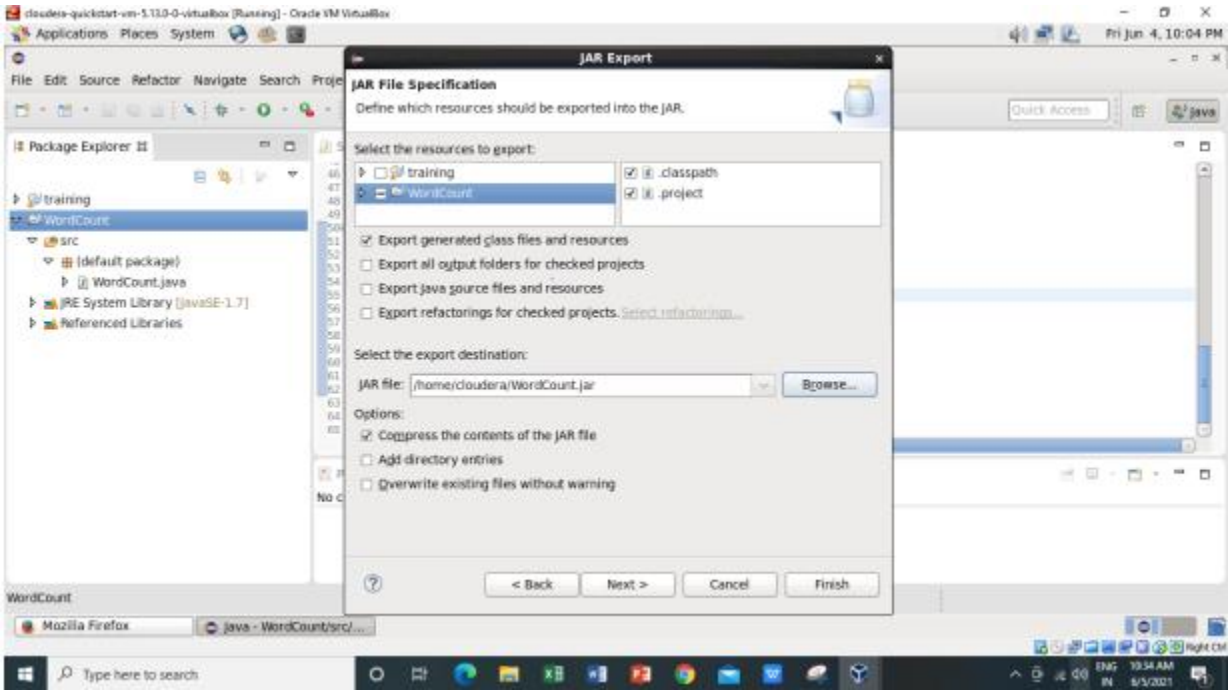
Name:
Roll No:


Export
Select
Choose export destination.
Select an export destination:
type filter text
General
Install
Java
JAR file
Javadoc
Runnable JAR file
Run/Debug
Tasks
Team
XML
< Back
Next >
Cancel
Finish
Package Explorer
training
WordCount
src
(default package)
WordCount.java
JRE System Library [JavaSE-1.7]
Referenced Libraries


Export
Select
Export resources into a JAR file on the local file system.
Select an export destination:
type filter text
General
Install
Java
JAR file
Javadoc
Runnable JAR file
Run/Debug
Tasks
Team
XML
< Back
Next >
Cancel
Finish
Package Explorer
training
WordCount
src
(default package)
WordCount.java
JRE System Library [JavaSE-1.7]
Referenced Libraries

Name:
Roll No:





4)Now checking the WordCount.jar file is created or not using –ls command

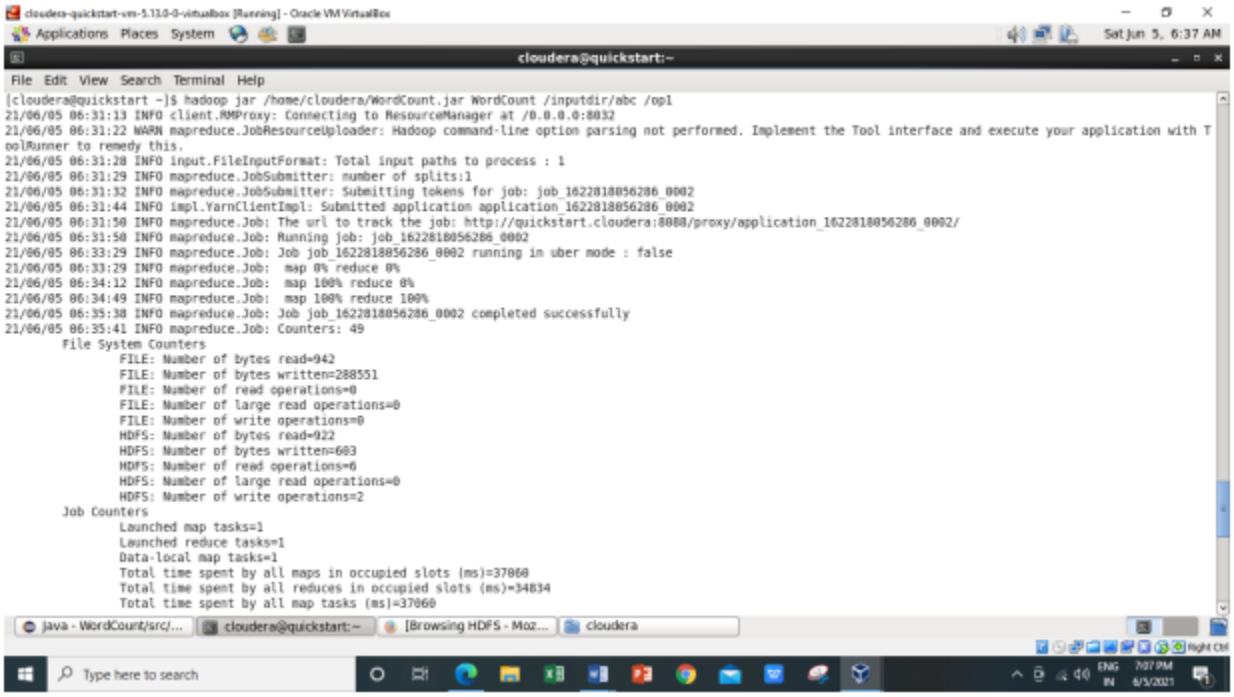
```
[cloudera@quickstart ~]$ ls
cloudera-manager  Desktop    Downloads  enterprise-deployment.json  kerberos  Music    Pictures  Templates  WordCount.jar
cm_api.py         Documents  eclipse    express-deployment.json     lib       parcels  Public    Videos     workspace
[cloudera@quickstart ~]$
```

Name:
Roll No:

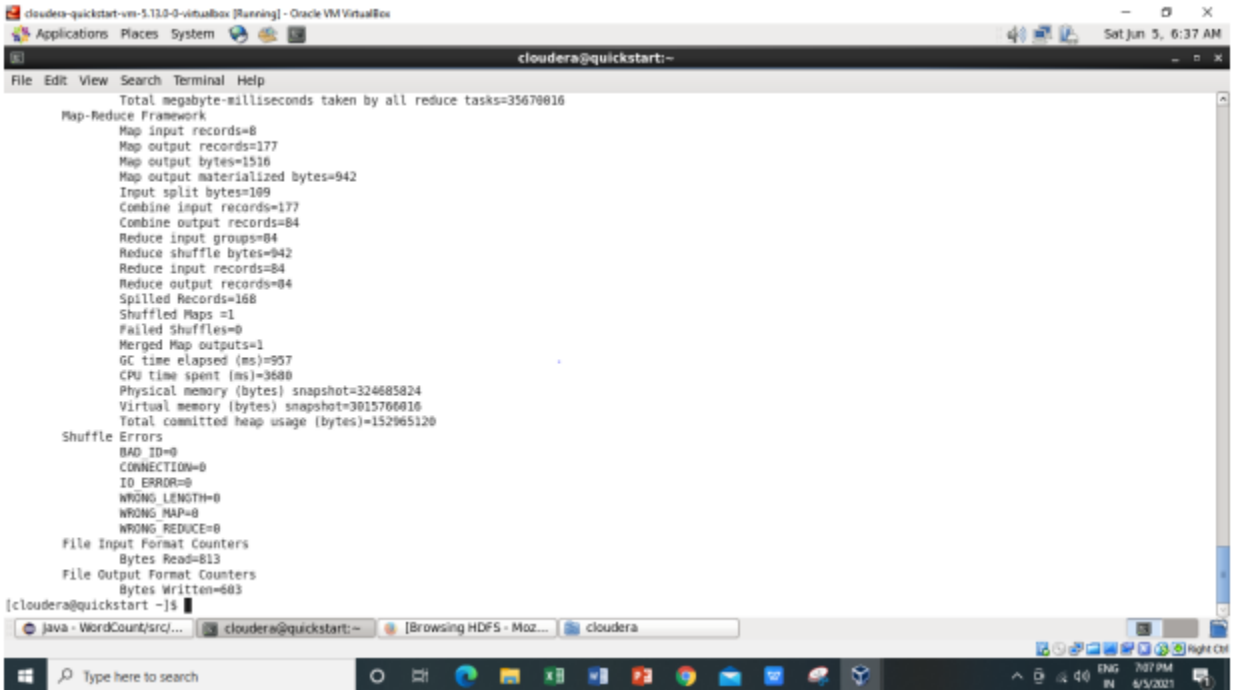
5) Running Mapreduce Program on Hadoop, syntax is hadoop jar jarFileName.jar ClassName /InputFileAddress /outputdir

i.e. hadoop jar /home/cloudera/WordCount.jar WordCount /inputdir/abc /op1

here I am using the same input file 'abc' which I have created earlier for WordCount example (Without Combiner**). For every execution of this program we need to delete the output directory or give a new name to the output directory every time.** So here I am giving the new name to the output directory as **'op1'.**



Name:
Roll No:



- As we can see from above image the the combiner input and output records coming out as,

  Combine input records=177

  Combine output records=84

- Earlier it was coming out as "zero" while executing WordCount (without combiner).

  Combine input records=0

  Combine output records=0

- And also here we are getting the Reduce Shuffle bytes as,

  Reduce shuffle bytes=942

  Earlier while executing WordCount (without combiner) it is coming out as,

  Reduce shuffle bytes=1876

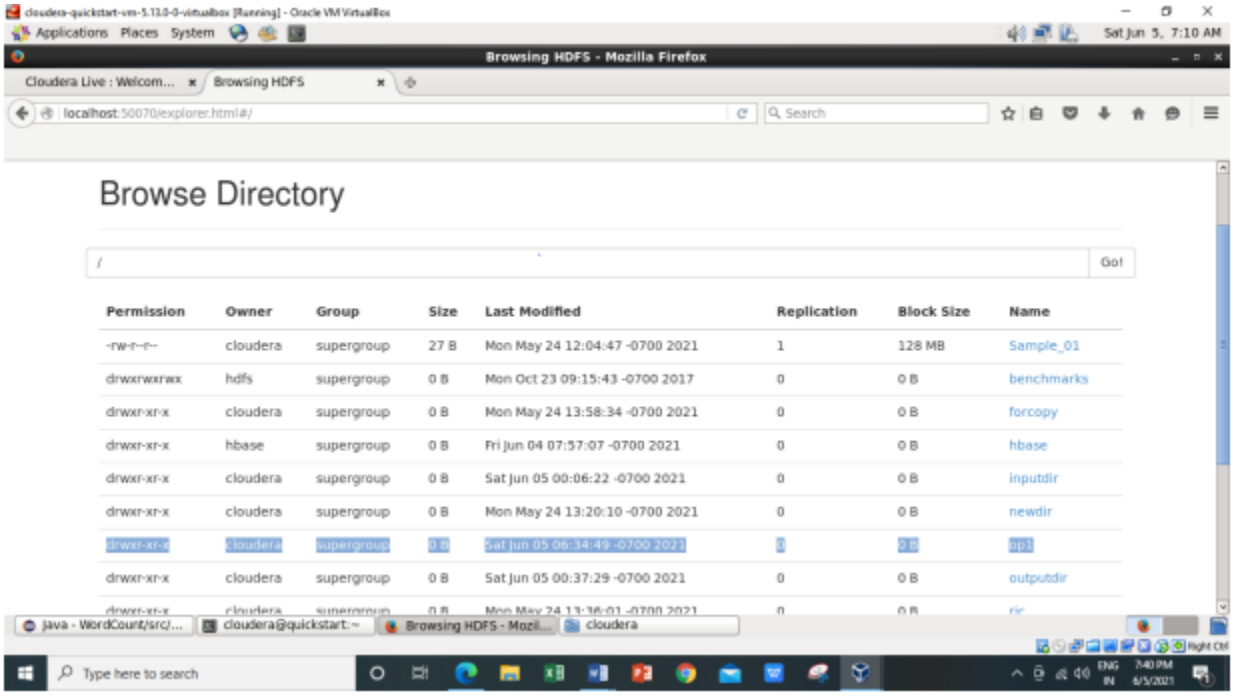
- So Combiner is used to save the Network Bandwidth. So for saving the Network bandwidth we make use of combiner. So instead of sending every word over the network what we do is we incorporate the logic of the reducer at the combiner side so that the less amount of information can be transmitted over the network.
- So when we are not using combiner 1876 bytes acting as an input for the reducer. And when we are making use of the combiner so 942 bytes acting as input for the reducer.

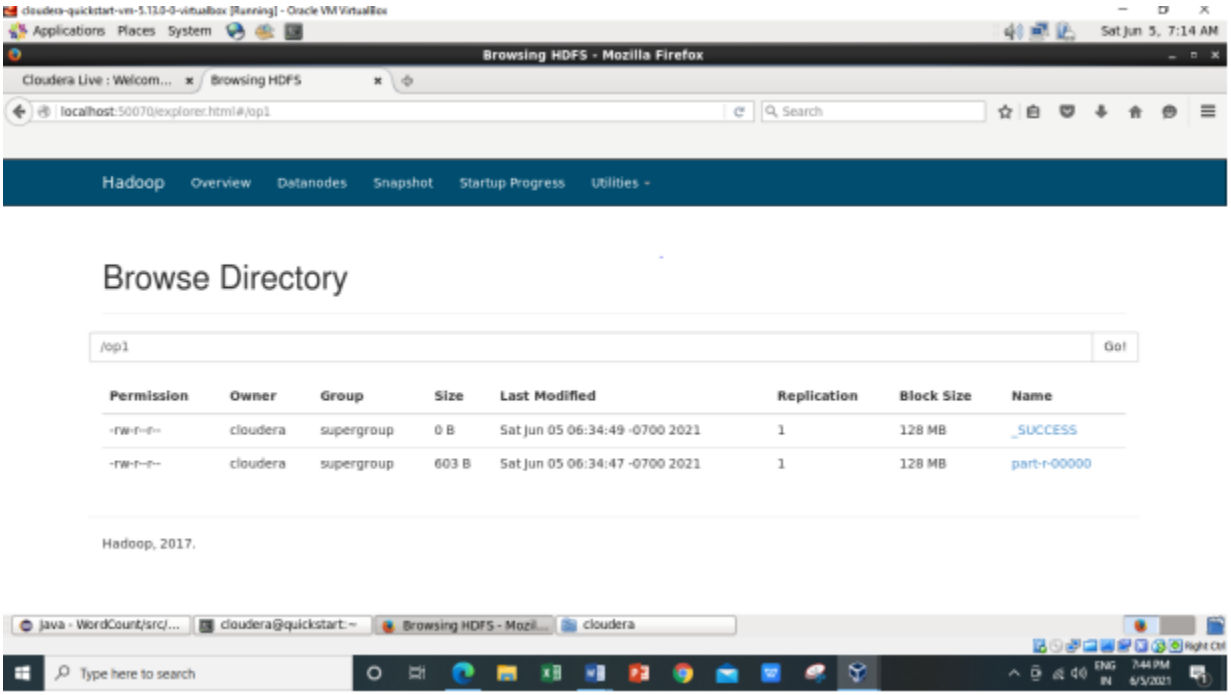
6) The same file can also be accessed using a browser.

Name:
Roll No:

Browse the Directory by

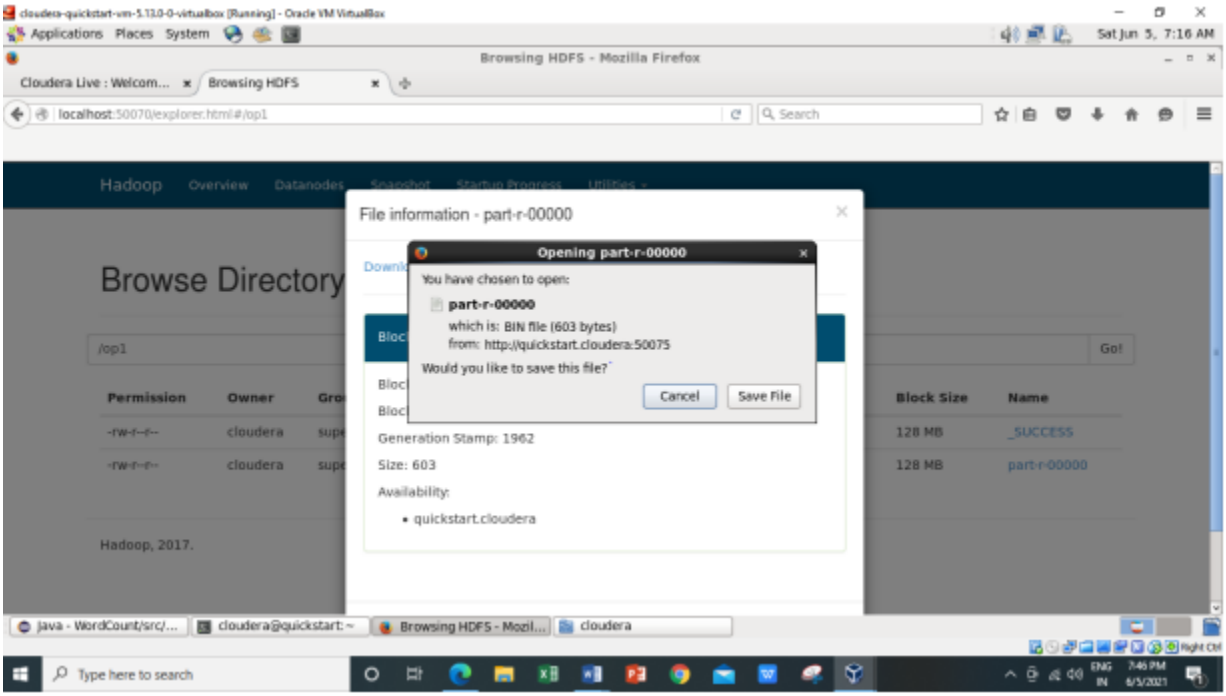
Hadoop->HDFS Namenode->Ultilities ->Browse the file system



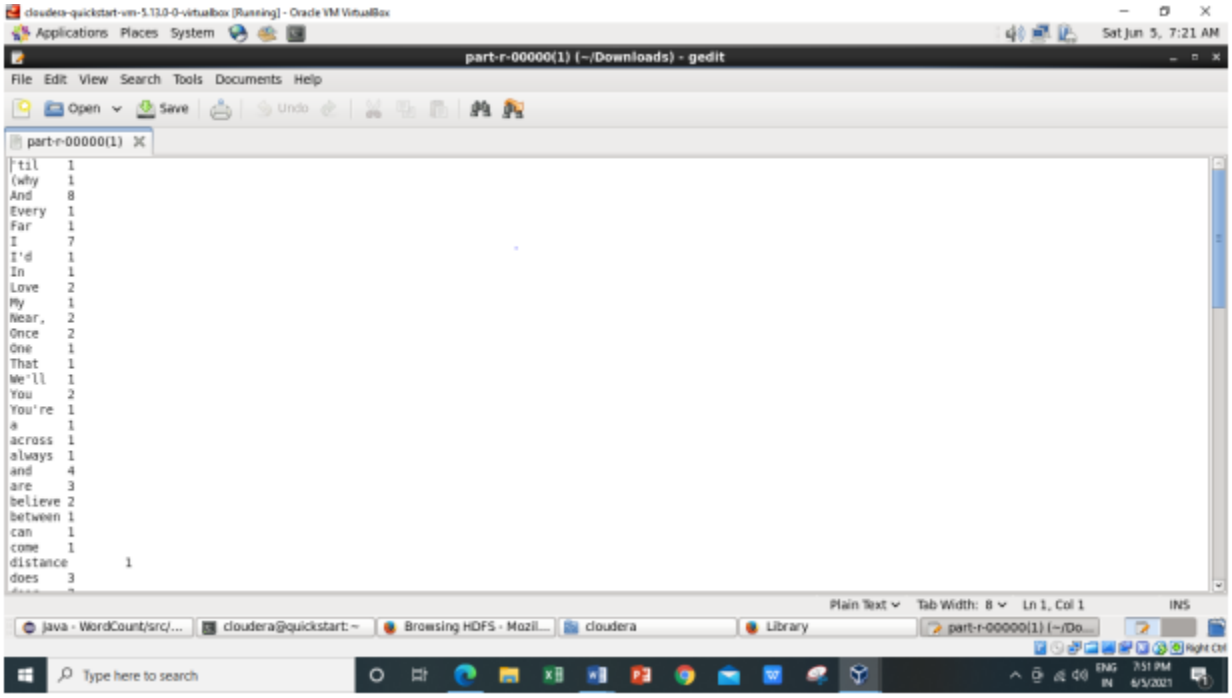


Now downloading the **part-r-00000** file.

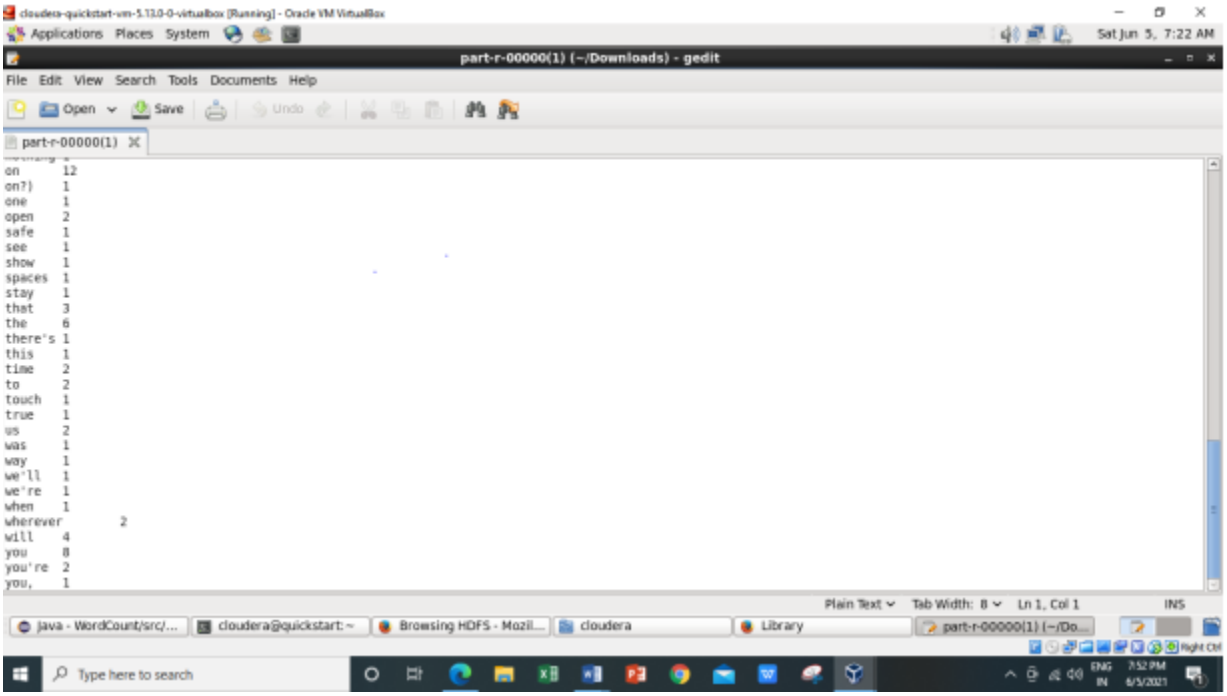
Name:
Roll No:



Inside the **part-r-00000** file it will have the same output as we are getting after executing using command **hadoop jar /home/cloudera/WordCount.jar WordCount /inputdir/abc /op1**



Name:
Roll No:


part-r-00000(1) (~/Downloads) - gedit
File Edit View Search Tools Documents Help
Open Save Undo
part-r-00000(1)
on 12
on?) 1
one 1
open 2
safe 1
see 1
show 1
spaces 1
stay 1
that 3
the 6
there's 1
this 1
time 2
to 2
touch 1
true 1
us 2
was 1
way 1
we'll 1
we're 1
when 1
wherever 2
will 4
you 8
you're 2
you, 1
Plain Text Tab Width: 8 Ln 1, Col 1 INS